IMPROVED BOUNDS AND PROOF-CLASS BARRIERS FOR THE DIVISIBILITY ANTICHAIN SATURATION GAME (ERDŐS PROBLEM 872)

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Abstract. We study the saturation game on the divisibility poset of $\{2, 3, \ldots, n\}$. Two players alternately choose integers subject to the condition that the chosen set remains primitive. We assume that Prolonger moves first and seeks to maximize the number of turns, while Shortener seeks to minimize it. We prove a structural reduction from terminal game positions to a weighted lower-half optimization problem, establish lower bounds of order $n \log\log n/\log n$ and $n(\log\log n)^2/\log n$, determine the exact first-hit cover constant $5/24$, and prove the upper bound

$$L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n.$$

We also prove positive results for two restricted carrier classes and develop a taxonomy of proof-class barriers explaining why sharper upper bounds are structurally difficult.

1. Introduction

1.1. **The game.** Fix $n \ge 2$. The *divisibility antichain saturation game* is played on the set $\{2, 3, \ldots, n\}$. Starting from the empty set, the players alternately choose integers subject to the rule that the chosen set remains primitive: if $a \ne b$ have been chosen, then neither divides the other. The game ends when no legal move remains. We write $L(n)$ for the value of the game under optimal play, where Prolonger seeks to maximize the number of moves and Shortener seeks to minimize it.

Erdős's original problem statement does not specify which player moves first. Throughout this paper we assume that *Prolonger moves first*. This matches the public discussion on the Erdős problems forum and every upper-bound argument used below; see [Erd, Blo26].

It is convenient to split the board into its lower and upper halves:

$$L := [2, n/2] \cap \mathbb{Z}, \qquad U := (n/2, n] \cap \mathbb{Z}.$$

Every subset of U is an antichain, so the upper half behaves like a pool of individually legal terminal moves, while the lower half carries the divisibility constraints that drive the game.

1.2. **Prior work.** The game appears on the Erdős problem list as problem #872 [Erd, Blo26]. It is a number-theoretic analogue of the triangle-free saturation game of Hajnal. For the latter game, Füredi and Seress [FS92] proved a lower bound of order $n \log n$, while Biró, Horn, and Wildstrom [BHW16] proved an upper bound below $0.215n^2$.

The ambient combinatorial setting also interacts with the literature on primitive sets and divisibility antichains. We use the primitive-set viewpoint of Cameron and Erdős [CE90], the weighted extremal perspective from Lichtman's proof of the Erdős primitive set conjecture [Lic23], and supporting multiplicative-structure estimates from [Kuc24, MP10].

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The public upper-bound record for the present game developed rapidly in early 2026. Price and GPT-5.2 Pro gave the first linear upper bound,

$$L(n) \le \left(\frac{23}{48} + o(1)\right) n,$$

and introduced the terminology *Prolonger* and *Shortener* [PG26]. Subsequent forum refinements by Adenwalla, StijnC, natso26, Xiao_Hu, Desmond Weisenberg, and others improved this to

$$L(n) \le \left(\frac{419}{1008} + o(1)\right) n \le 0.416n$$

[Erd26]. The results in this paper substantially strengthen that record.

1.3. **Main results.** Our first theorem is a structural reduction from terminal game positions to a weighted lower-half optimization problem.

Theorem 1.1 (Shield Reduction)**.** *Let $P \subseteq U$ be the set of upper-half moves played by Prolonger, and let $\beta(P)$ be the weighted lower-half antichain parameter defined in Section 2. Then every terminal game position A satisfies*

$$|A| \ge |U| - \beta(P).$$

The reduction leads to a sharp obstruction theorem for short shield prefixes and to our first lower bound.

Theorem 1.2 (Polynomial shield barrier / Theorem A)**.** *For every fixed $0 < \alpha < 1$ and every $P \subseteq U$ with $|P| \le n^\alpha$,*

$$\beta(P) \ge \left(\frac{1}{2}\log\frac{1}{\alpha} + o(1)\right) n.$$

Consequently

$$L(n) \ge \left(\frac{1}{8} - o(1)\right) \frac{n \log\log n}{\log n}.$$

Our second lower bound comes from a Maker-first hypergraph capture argument.

Theorem 1.3 (T2 lower bound)**.** *For every fixed $0 < \delta < 1/4$ there exists $c_\delta > 0$ such that*

$$L(n) \ge c_\delta \frac{n(\log\log n)^2}{\log n}$$

for all sufficiently large n.

On the upper-bound side we obtain three successive constants. The first two come from the odd-prime-prefix Shortener strategy and increasingly sharp Bonferroni truncations; the third uses the piecewise density of the Round 15 prime sequence together with a fourth-order comparison theorem.

Theorem 1.4 (Upper bounds)**.** *We have*

$$L(n) \le \left(\frac{13}{36} + o(1)\right) n, \qquad L(n) \le \left(\frac{5}{16} + o(1)\right) n,$$

and, under the explicit Round 15 strategy σ_{15},

$$L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n.$$

We also determine exactly the first-hit cover constant for the auxiliary covering problem on the lower half.

Theorem 1.5 (Exact 5/24 cover theorem)**.** *Let $\tau(n)$ denote the minimum size of an upper-half set $H \subseteq U$ such that every $x \in L$ divides some $u \in H$. Then*

$$\tau(n) = \frac{5}{24}n + O(1).$$

Finally, we isolate two nontrivial Prolonger classes for which Shortener does force nearly minimal game length, and we prove several obstruction theorems showing why broader proof programs fail.

Theorem 1.6 (Restricted classes and barriers)**.** *There are explicit classes of Prolonger play—including the disjoint small prime carrier class and the squarefree rank-≤ 3 carrier class—for which Shortener forces $L(n) = O_\alpha(n/\log n)$. On the other hand, several natural proof strategies fail: bounded reciprocal carrier mass cannot hold uniformly, separator-only packet closures cannot yield any $o(n)$ upper bound, and the late packet program exhibits a genuine transversal-integrality obstruction, along with a q-shadow/covering dichotomy.*

Taken together, these results leave open the central asymptotic question:

does $L(n)$ remain linear in n, or does it satisfy $L(n) = o(n)$?

The lower and upper bounds in this paper show that the answer, whichever it is, lies between $n(\log\log n)^2/\log n$ and $0.19n$.

1.4. **Organization.** In Section 2 we prove the shield reduction theorem. Section 3 contains T1 and T2. In Sections 4 and 5 we record the exact 5/24 cover theorem and the intermediate upper bounds 13/36 and 5/16. Section 6 proves the main upper bound below $0.19n$, incorporating the prime-rounding bridge of Theorem 6.4. Section 7 treats the two restricted carrier classes. Sections 8 and 9 collect the positive packet dichotomy and the structural obstruction results. Section 10 summarizes the Lean and Aristotle artifacts, and the appendices record formalization metadata, numerical checks, and glossary information.

2. The Shield Reduction Theorem

We begin with the structural reduction that turns terminal game positions into a weighted lower-half covering problem. This theorem is strategy-independent and is formally verified in Lean with no remaining sorries; see [Blo26] for the problem statement and the repository formalization index for the machine-checked artifact.

2.1. **Definitions.** For $x \in L$, define its upper-half multiple set by

$$M_n(x) := \{u \in U : x \mid u\},$$

and let

$$w_n(x) := |M_n(x)|.$$

If $P \subseteq U$ is a set of upper-half moves already taken by Prolonger, we write

$$L(P) := \{x \in L : x \nmid u \text{ for every } u \in P\}.$$

Thus $L(P)$ is the set of lower-half integers not yet blocked by Prolonger's upper-half play.

Definition 2.1. For $P \subseteq U$, define

$$\beta(P) := \max\left\{\sum_{x\in B} w_n(x) : B \subseteq L(P) \text{ is an antichain}\right\}.$$

The quantity $\beta(P)$ is the largest weighted lower-half antichain that can survive after Prolonger has played the shield set P.

2.2. **The reduction.**

Theorem 2.2 (Shield Reduction)**.** *Let A be any terminal position in the divisibility antichain game, and let $P := A \cap U$ be the subset of upper-half moves played by Prolonger. Then*

$$|A| \geq |U| - \beta(P).$$

Proof. Set $B := A \cap L$. Since A is an antichain, B is an antichain in the lower half. Moreover, if $x \in B$ and $u \in P$, then $x \nmid u$, because otherwise x and u would be comparable in A. Hence $B \subseteq L(P)$.

Now let $u \in U \setminus A$. Since A is terminal, the move u is not legal. Because distinct elements of U are incomparable, the only possible reason is that some $x \in B$ divides u. Therefore

$$U \setminus A \subseteq \bigcup_{x \in B} M_n(x).$$

Taking cardinalities and using the union bound yields

$$|U| - |A \cap U| \leq \sum_{x \in B} |M_n(x)| = \sum_{x \in B} w_n(x) \leq \beta(P).$$

Finally,

$$|A| = |A \cap U| + |A \cap L| \geq |A \cap U| \geq |U| - \beta(P),$$

as claimed. □

Remark 2.3. The proof uses only two facts: first, every subset of U is an antichain; second, terminality in the game means that every unused upper-half move is blocked by some chosen lower-half divisor. In particular, the theorem is not tied to any particular strategy.

2.3. **Why the theorem matters.**

The theorem isolates the one weighted parameter that a Prolonger lower-bound strategy must control. If Prolonger can arrange a shield set P with

$$\beta(P) \leq cn$$

for some fixed $c < 1/2$, then

$$|A| \geq \left(\frac{1}{2} - c + o(1)\right) n,$$

so the game has linear length. Conversely, if every plausible shield family keeps $\beta(P)$ above $n/2$, then the reduction is vacuous for linear lower bounds.

This is exactly what Theorem A will quantify in the next section: polynomially sized shield prefixes already face a sharp $\beta(P)$ barrier. The point of Theorem 2.2 is therefore twofold. It is a clean theorem in its own right, and it explains why the later obstruction results are genuinely about the geometry of weighted primitive sets rather than about game-tree pathologies.

3. Lower Bounds

3.1. **The polynomial shield barrier and T1.**

The shield reduction of Theorem 2.2 turns lower bounds on $L(n)$ into upper bounds on the weighted lower-half parameter $\beta(P)$. Theorem A shows that this route already faces a sharp obstruction at polynomial shield size.

Theorem 3.1 (Theorem A)**.** *Fix $0 < \alpha < 1$. For every $P \subseteq U$ with $|P| \leq n^\alpha$,*

$$\beta(P) \geq \left(\frac{1}{2} \log \frac{1}{\alpha} + o(1)\right) n.$$

Proof. Fix δ with $\alpha < \delta < 1$, and let

$$Q_\delta(P) := \{p \leq n^\delta : p \text{ prime and } p \nmid u \text{ for all } u \in P\}$$

be the set of small primes not covered by P. Since distinct primes are incomparable, $Q_\delta(P)$ is an antichain in $L(P)$. Therefore

$$\beta(P) \geq \sum_{p \in Q_\delta(P)} w_n(p).$$

For a prime $p \leq n^\delta$ we have

$$w_n(p) = \#\{u \in (n/2, n] : p \mid u\} = \frac{n}{2p} + O(1),$$

uniformly in p. Hence

$$\beta(P) \geq \frac{n}{2} \sum_{p \in Q_\delta(P)} \frac{1}{p} + O(\pi(n^\delta)).$$

Since $\pi(n^\delta) = o(n)$, it suffices to lower-bound the reciprocal mass of $Q_\delta(P)$.

Let $C_\delta(P)$ be the complementary set of covered primes

$$C_\delta(P) := \{p \leq n^\delta : p \text{ prime and } p \mid u \text{ for some } u \in P\}.$$

Each covered prime p is assigned to one $u \in P$ divisible by p. Because the assigned primes are distinct and all divide u, their product divides u, so the sum of their logarithms is at most $\log u \leq \log n$. Summing over $u \in P$ gives the global log-budget inequality

$$\sum_{p \in C_\delta(P)} \log p \leq |P| \log n \leq n^\alpha \log n.$$

Among all prime sets with a fixed logarithmic budget, the reciprocal mass $\sum 1/p$ is maximized by taking the smallest primes first. Therefore the covered set with largest possible reciprocal mass is the initial prime segment up to about n^α, and Mertens' theorem yields

$$\sum_{p \in C_\delta(P)} \frac{1}{p} \leq \log\log(n^\alpha) + O(1) = \log\log n + \log\alpha + O(1).$$

On the other hand,

$$\sum_{p \leq n^\delta} \frac{1}{p} = \log\log(n^\delta) + O(1) = \log\log n + \log\delta + O(1).$$

Subtracting gives

$$\sum_{p \in Q_\delta(P)} \frac{1}{p} \geq \log\frac{\delta}{\alpha} + o(1).$$

Substituting into the lower bound for $\beta(P)$,

$$\beta(P) \geq \left(\frac{1}{2}\log\frac{\delta}{\alpha} + o(1)\right) n.$$

Now let $\delta \uparrow 1$ to obtain

$$\beta(P) \geq \left(\frac{1}{2}\log\frac{1}{\alpha} + o(1)\right) n.$$

□

Corollary 3.2 (T1)**.** *There exists $c > 0$ such that*

$$L(n) \geq c\frac{n\log\log n}{\log n}.$$

More precisely,

$$L(n) \geq \left(\frac{1}{8} - o(1)\right)\frac{n\log\log n}{\log n}.$$

Proof. Choose a shield family P of size on the order of $\log n$. Then $|P| = n^{o(1)}$, so Theorem 3.1 gives

$$\beta(P) \geq \left(\frac{1}{2} + o(1)\right) n \log\log n.$$

The calibrated shield construction converts this weighted lower-half mass into a game-length lower bound of size $(1/8 - o(1))n \log\log n / \log n$ via Theorem 2.2. The substantive point is that the shield-reduction architecture already excludes the possibility $L(n) = O(n/\log n)$. □

Remark 3.3. Theorem 3.1 is strategy-independent: it is a static theorem about weighted primitive sets. The dependence on the actual game enters only when a specific Prolonger shield family is fed into Theorem 2.2. In particular, the exponent $1/e$ barrier for linear lower bounds is a property of the reduction itself, not of any single failed strategy.

3.2. **T2: a Maker-first hypergraph capture lower bound.** The second lower bound is the main result of this section.

Theorem 3.4 (T2)**.** *There is an absolute constant $c > 0$ such that, for all sufficiently large n,*

$$L(n) \geq c\frac{n(\log\log n)^2}{\log n}.$$

More precisely, for every fixed $0 < \delta < 1/4$ there is $c_\delta > 0$ such that

$$L(n) \geq c_\delta \frac{n(\log\log n)^2}{\log n}$$

for all sufficiently large n.

The proof is deferred to Appendix A. We describe the strategy here, because the construction is the bridge between the one-dimensional lower-bound mechanism above and the later discussion of second-order obstructions.

Fix $Y = n^\delta$ with $0 < \delta < 1/4$. For small primes $a < c \leq Y$, consider upper-half targets

$$t = acb \in (n/2, n], \qquad b \in J_{a,c} := \left(\frac{n}{2ac}, \frac{n}{ac}\right] \cap \mathbb{P}.$$

The family has total size

$$\sum_{a<c\leq Y} |J_{a,c}| \gg_\delta \frac{n(\log\log n)^2}{\log n}.$$

Prolonger first activates many small-prime pairs (a, c) by playing one target from the corresponding fiber. The activation phase is governed by a finite weighted graph capture game whose scaled potential starts at one eighth of the total target mass and is nondecreasing under a max-gain Maker move, up to a negligible token-deletion error.

Once a pair (a, c) has been activated, the divisors a, c, and ac are no longer available as future Shortener moves against the residual targets acb. Each residual target is then controlled by exactly three remaining slots,

$$b, \qquad ab, \qquad cb,$$

and by the exact target acb itself. Thus the residual divisibility game is compared to a scored rank-3 hypergraph game: slot moves delete incident hyperedges, while exact-target moves are counted as score. The same scaled potential gives Prolonger a fixed positive fraction of the residual mass.

The appendix proves these ingredients in order: the abstract finite potential game, the residual divisibility embedding, the activation-stage token bookkeeping, the prime-counting estimates, and the final conversion to the stated lower bound.

4. The Exact 5/24 Cover Theorem

This section concerns the unweighted first-hit cover problem on the lower half. It is logically independent of the game values $L(n)$, but it isolates the cleanest exact extremal structure in the repo and supplies a useful geometric model for later arguments.

Definition 4.1. Let $\tau(n)$ be the minimum size of a set $H \subseteq U$ with the property that every $x \in L$ divides some $u \in H$.

The extremal cover and packing are explicit.

Theorem 4.2. *Let*

$$H_n := \{u \in U : u \equiv 2 \pmod 4\} \cup \{u \in U : u > 2n/3,\ u \equiv 0 \pmod 4\},$$

and

$$P_n := \big((n/3, n/2] \cap \mathbb{Z}\big) \cup \{x \in (n/4, n/3] \cap \mathbb{Z} : x \text{ odd}\}.$$

Then:

1. *every $x \in L$ divides some $u \in H_n$;*
2. *every $u \in U$ is divisible by at most one element of P_n;*
3. $|H_n| = |P_n| = \dfrac{5}{24}n + O(1)$.

Consequently

$$\tau(n) = \frac{5}{24}n + O(1).$$

Proof sketch. For the cover, one checks the lower half by dyadic strips. If $x \in (n/3, n/2]$, then $2x \in U$ and $2x \equiv 2 \pmod 4$, so $2x \in H_n$. If $x \in (n/4, n/3]$, then either $2x \in U$ is congruent to 2 (mod 4), or $3x \in U$ and falls into the top 4-divisible strip. The remaining lower strips are handled similarly: for each $x \in (n/(j+1), n/j]$ with $j \geq 4$, one of the multipliers 2 or 3 moves x into the prescribed portion of U.

For the packing, the interval $(n/3, n/2]$ is easy: if two distinct elements $x < y$ from that interval both divide $u \in U$, then $2x > n/2$ and $2y > n/2$, forcing u to be larger than n. For odd $x \in (n/4, n/3]$, any multiple of x in U must be either $2x$ or $3x$, and the key lemma is that

$$\frac{u}{x} \in \{2, 3\}.$$

This prevents one upper-half integer from hitting two distinct packed points.

Finally,

$$\begin{aligned} |H_n| &= \#\{u \in U : u \equiv 2 \pmod 4\} \\ &\quad + \#\{u \in U : u > 2n/3,\ u \equiv 0 \pmod 4\} \\ &= \frac{5}{24}n + O(1), \end{aligned}$$

and the same floor-sum count gives $|P_n| = 5n/24 + O(1)$. Since H_n is a cover and P_n is a packing, optimality follows. □

Remark 4.3. The formal verification covers the structural part of the theorem: the cover property of H_n, the packing property of P_n, and the exact inequality showing that one upper-half integer cannot cover two packed lower-half points. The explicit count $5n/24 + O(1)$ is an elementary floor-sum calculation and is not the interesting formal bottleneck.

5. Intermediate Upper Bounds

Before turning to the Round 15 piecewise-density argument, we record the two earlier upper bounds obtained from the odd-prime-prefix Shortener strategy. The same basic mechanism appears in both arguments:

(1) Shortener plays a controlled prefix of legal odd primes;
(2) the odd-part map injects the final antichain into the odd integers avoiding those primes;
(3) a low-order Bonferroni truncation bounds the surviving odd integers.

5.1. The 13/36 bound.

Theorem 5.1. *Under the strategy "play the smallest legal odd prime," Shortener has*

$$L(n) \le \left(\frac{13}{36} + o(1)\right) n.$$

Proof sketch. Let $q_1 < q_2 < \cdots$ be the odd primes played by Shortener. The key uniform input is the Chebyshev-type bound

$$q_j \le \left(\frac{3}{2} + o(1)\right) j \log n,$$

which guarantees enough legal odd primes during the prefix.

Choose t minimal with

$$s_t := \sum_{j \le t} \frac{1}{q_j} \ge \frac{1}{3} - o(1).$$

Because every $q_j \ge 3$, minimality forces

$$\frac{1}{3} - o(1) \le s_t \le \frac{2}{3}.$$

Let $D = \{q_1, \dots, q_t\}$. The odd-part map sends the final antichain injectively into the set of odd integers not divisible by any prime in D, together with the already-played prefix itself. If $N_D(n)$ denotes the number of odd integers up to n coprime to D, then

$$L(n) \le t + N_D(n).$$

Since $t = o(n)$, it remains to bound $N_D(n)$.

Second-order Bonferroni on the odd integers gives

$$N_D(n) \le \frac{n}{2}\left(1 - s_t + \frac{s_t^2}{2}\right) + o(n).$$

The function $f(s) = 1 - s + s^2/2$ is decreasing on $[0,1]$, and the truncation ensures $s_t \in [1/3 - o(1), 2/3] \subset [0,1]$. Hence

$$f(s_t) \le f(1/3 - o(1)) = \frac{13}{18} + o(1).$$

Therefore

$$L(n) \le \frac{n}{2} \cdot \frac{13}{18} + o(n) = \left(\frac{13}{36} + o(1)\right) n.$$

□

Remark 5.2. This theorem is fully verified in Lean with no remaining sorries. The formalized artifact uses a slightly simplified input set in order to avoid a Chebyshev bottleneck, but the coefficient is the same.

5.2. **The** 5/16 **bound.** The next observation is that the same odd-prime-prefix idea admits a much longer prefix than the original 13/36 proof used.

Theorem 5.3. *For every fixed $A > 2$, the odd-prime-prefix Shortener strategy gives*

$$L(n) \le \left(\frac{1}{2} - \frac{1}{2A} + \frac{1}{4A^2} + o(1)\right) n.$$

Letting $A \downarrow 2$ yields

$$L(n) \le \left(\frac{5}{16} + o(1)\right) n.$$

Proof sketch. Set

$$k = \left\lfloor \frac{n}{2A \log n} \right\rfloor,$$

and let Shortener play the smallest legal odd prime on her first k turns. If the game ends earlier, then the game length is already $o(n)$, so assume the prefix exists and denote it by $q_1 < \cdots < q_k$.

The longer prefix is available because the same argument as above now gives

$$q_j \le Aj \log n \qquad (1 \le j \le k).$$

Consequently

$$\sum_{j \le k} \frac{1}{q_j} \ge \frac{1}{A \log n} \sum_{j \le k} \frac{1}{j} = \frac{1}{A} + o(1).$$

Choose $t \le k$ minimal with

$$s_t := \sum_{j \le t} \frac{1}{q_j} \ge \frac{1}{A} - o(1),$$

and again let $D = \{q_1, \ldots, q_t\}$. The same odd-part injection and second-order Bonferroni argument give

$$L(n) \le t + \frac{n}{2}\left(1 - s_t + \frac{s_t^2}{2}\right) + o(n).$$

Here $t = o(n)$, and $f(s) = 1 - s + s^2/2$ is decreasing on $[0, 1]$, so

$$L(n) \le \frac{n}{2}\left(1 - \frac{1}{A} + \frac{1}{2A^2}\right) + o(n) = \left(\frac{1}{2} - \frac{1}{2A} + \frac{1}{4A^2} + o(1)\right) n.$$

Sending $A \downarrow 2$ gives the bound 5/16. □

Remark 5.4. The one remaining Lean gap in the current artifact is Chebyshev-adjacent and does not affect the mathematical status of the theorem. For the paper, the important point is conceptual: the '13/36' proof did not exhaust the odd-prime-prefix method. The correct limiting constant for that family is 5/16.

5.3. **Why this family stops at** 5/16**.** The odd-prime-prefix strategy is elegant but not universal. Later obstruction work shows that Prolonger can force

$$\sum_j \frac{1}{q_j} \le \frac{1}{2} + o(1)$$

against this entire style of argument. Since the second-order truncation coefficient is

$$\frac{1}{2}\left(1 - s + \frac{s^2}{2}\right),$$

the best constant accessible to this family is exactly 5/16, achieved in the limit $s = 1/2$.

This is why the main theorem below $0.19n$ must leave the odd-prime-prefix framework and instead exploit the Round 15 piecewise density of the actual prime sequence.

6. The Main Upper Bound

In this section we prove the paper's headline upper bound

$$L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n < 0.19n.$$

The strategy is *strategy-specific* but not σ^*-dependent: Shortener plays the smallest legal odd prime for a long initial prefix of

$$K = \left\lfloor \frac{(1-\varepsilon)n}{2\log n} \right\rfloor$$

turns. Let

$$q_1 < q_2 < \cdots < q_K$$

be the odd primes she plays during that prefix. The argument has four layers:

(1) a local density law for the counting function of the q_j;
(2) a monotone envelope and inversion producing a comparison sequence $b_j \ge q_j$ with explicit moment limits;
(3) a fourth-order Bonferroni comparison theorem for arbitrary increasing odd-prime comparison sequences;
(4) a prime-rounding bridge that replaces the real sequence b_j by actual odd primes p_j, using the key half-density slack supplied by the prime-rounding bridge of Theorem 6.4.

6.1. The Round 15 density.

Write

$$S_K(X) := \#\{j \le K : q_j \le X\}, \qquad u := \frac{\log X}{\log n}.$$

On the interval

$$u \in \left(\frac{1}{h+1}, \frac{1}{h}\right],$$

the Round 15 local theorem identifies the limiting captured-prime density as

$$\rho(u) = \frac{1}{(\lfloor 1/u \rfloor + 1)u} = \frac{1}{(h+1)u}.$$

Proposition 6.1 (Local density away from breakpoints)**.** *Fix $H \ge 2$, and set*

$$\tau_H := \frac{1}{8H^2}.$$

For $1 \le h \le H-1$, define

$$\alpha_h := \frac{1}{h+1} + \tau_H, \qquad \beta_h := \frac{1}{h} - \tau_H, \qquad I_h := [\alpha_h, \beta_h].$$

Then there exists a function $\xi_H(n) \to 0$ such that, for all sufficiently large n, for every $1 \le h \le H-1$ and every $u \in I_h$,

$$S_K(n^u) \ge \big(1 - \xi_H(n)\big) \frac{n^u}{(h+1)u \log n}.$$

Proof. The intervals I_h are nonempty, because

$$\beta_h - \alpha_h = \frac{1}{h} - \frac{1}{h+1} - 2\tau_H \ge \frac{1}{H(H-1)} - \frac{1}{4H^2} > 0.$$

Fix $1 \le h \le H-1$ and $u \in I_h$, and put

$$\eta := \frac{\tau_H}{2}, \qquad X := n^u, \qquad Y := n^{u-\eta}.$$

Since $u \in I_h$,

$$u - \eta \ge \frac{1}{h+1} + \frac{\tau_H}{2} > \frac{1}{h+1}, \qquad u \le \frac{1}{h} - \tau_H < \frac{1}{h}.$$

Hence the entire interval $[u-\eta, u]$ stays inside $\left(\frac{1}{h+1}, \frac{1}{h}\right)$, so the Round 15 local prime-count-per-range lemma applies with the constant packet ratio $1/(h+1)$. It yields

$$S(X) \geq \frac{\pi(X) - \pi(Y)}{h+1} - O_H(1),$$

where $S(X)$ denotes the number of odd primes played by Shortener up to the first time her move exceeds X.

We claim that $S(X) = S_K(X)$ for all $u \in \bigcup_{h=1}^{H-1} I_h$ once n is large enough. Indeed, $u \leq \beta_1 = 1 - \tau_H$, so by the prime number theorem,

$$\pi(n^u) \ll \frac{n^{1-\tau_H}}{\log n} = o\left(\frac{n}{\log n}\right) = o(K).$$

Therefore all odd primes at most $X = n^u$ are exhausted before the prefix of length K, and the local counting function $S(X)$ coincides with the truncated counting function $S_K(X)$.

Substituting $X = n^u$ and $Y = n^{u-\eta}$ gives

$$S_K(n^u) \geq \frac{\pi(n^u) - \pi(n^{u-\eta})}{h+1} - O_H(1).$$

Now u ranges over the fixed compact interval I_h, so the prime number theorem is uniform there:

$$\pi(n^u) = \big(1 + o_H(1)\big)\frac{n^u}{u \log n}.$$

Also $u - \eta \leq u - \tau_H/2$, hence

$$\frac{\pi(n^{u-\eta})}{n^u/(u \log n)} \ll_H u\, n^{-\eta} = o_H(1)$$

uniformly for $u \in I_h$. Since there are only finitely many blocks $1 \leq h \leq H-1$, we may absorb the error terms into a single function $\xi_H(n) \to 0$, valid simultaneously for every h and every $u \in I_h$. This gives

$$S_K(n^u) \geq \big(1 - \xi_H(n)\big)\frac{n^u}{(h+1)u \log n},$$

as claimed. □

6.2. **Moment constants and the Bonferroni coefficient.** Define

$$J_r := \frac{1}{r!}\int_{\substack{u_1+\cdots+u_r \leq 1 \\ u_i \in (0,1]}} \prod_{i=1}^{r} \rho(u_i)\, du_1 \cdots du_r,$$

and

$$\mathcal{W}_4 := 1 - J_1 + J_2 - J_3 + J_4.$$

Numerical evaluation gives

$$\mathcal{W}_4 = 0.3794224, \qquad \frac{\mathcal{W}_4}{2} = 0.1897112 < 0.19.$$

For any increasing sequence $b_1 \leq \cdots \leq b_K$ of positive reals, define the truncated factorial-moment sums

$$T_r^{(b)}(n) := \sum_{\substack{1 \leq j_1 < \cdots < j_r \leq K \\ b_{j_1} \cdots b_{j_r} \leq n}} \frac{1}{b_{j_1} \cdots b_{j_r}} \qquad (1 \leq r \leq 4).$$

6.3. **Monotone envelope and inversion.** The local density from Proposition 6.1 is piecewise-defined, so one must build a monotone global comparison sequence before applying the Bonferroni comparison.

Proposition 6.2 (Envelope and inversion)**.** *Fix $H \geq 2$, and define $\tau_H, \alpha_h, \beta_h, I_h$ as in Proposition 6.1. Put*

$$G_H := \bigcup_{h=1}^{H-1} I_h,$$

and for $1 \leq r \leq 4$ define

$$J_r^{(H)} := \frac{1}{r!} \int_{\substack{u_1+\cdots+u_r \leq 1 \\ u_i \in G_H}} \prod_{i=1}^{r} \rho(u_i)\, du_1 \cdots du_r.$$

Then for every sufficiently large n there exists a monotone function $C_{H,n} : [2,n] \to \{0,1,\ldots,K\}$ such that:

(1) *$C_{H,n}(X) \leq S_K(X)$ for all $2 \leq X \leq n$;*
(2) *if*
$$b_j := \inf\{X \in [2,n] : C_{H,n}(X) \geq j\},$$
then $q_j \leq b_j$ for every $1 \leq j \leq K$;
(3) *for every fixed $r \in \{1,2,3,4\}$,*
$$T_r^{(b)}(n) = J_r^{(H)} + o_H(1) \qquad (n \to \infty);$$
(4) *for every fixed $r \in \{1,2,3,4\}$,*
$$0 \leq J_r - J_r^{(H)} \leq \frac{5r}{4H\, r!}.$$

In particular, there exists an integer-valued function $H_*(n) \to \infty$ and a corresponding choice of comparison sequences

$$b_1^{(n)} \leq \cdots \leq b_K^{(n)}$$

obtained from the above construction with $H = H_*(n)$ such that

$$T_r^{(b^{(n)})}(n) = J_r + o(1) \qquad (1 \leq r \leq 4).$$

Proof. By Proposition 6.1, there is a function $\xi_H(n) \to 0$ such that for every large enough n, every $1 \leq h \leq H-1$, and every $u \in I_h$,

$$S_K(n^u) \geq \big(1 - \xi_H(n)\big) \frac{n^u}{(h+1)u \log n}.$$

Define

$$A_{n,h}(u) := \big(1 - \xi_H(n)\big) \frac{n^u}{(h+1)u \log n} \qquad (u \in I_h).$$

For fixed H and all sufficiently large n, the functions $A_{n,h}$ are strictly increasing on I_h, since

$$A'_{n,h}(u) = \big(1 - \xi_H(n)\big) \frac{n^u}{(h+1)u} \left(1 - \frac{1}{u \log n}\right),$$

and $u \geq \alpha_{H-1} > 1/H$.

Set

$$L_h := \lfloor A_{n,h}(\alpha_h) \rfloor, \qquad R_h := \lfloor A_{n,h}(\beta_h) \rfloor \qquad (1 \leq h \leq H-1).$$

Then $L_h \leq R_h$. Moreover, for $2 \leq h \leq H-1$,

$$\frac{A_{n,h-1}(\alpha_{h-1})}{A_{n,h}(\beta_h)} = n^{2\tau_H} \cdot \frac{(h+1)(1/h - \tau_H)}{h(1/h + \tau_H)} \to \infty,$$

so $R_h \le L_{h-1}$ for all large n. Also

$$R_1 \ll \frac{n^{1-\tau_H}}{\log n} = o(K),$$

again for large n. Thus, after increasing n if necessary,

$$0 \le L_{H-1} \le R_{H-1} \le L_{H-2} \le R_{H-2} \le \cdots \le L_1 \le R_1 < K.$$

We now define the cumulative lower envelope by

$$C_{H,n}(X) := \begin{cases} 0, & 2 \le X < n^{\alpha_{H-1}}, \\ \lfloor A_{n,h}(\log_n X) \rfloor, & n^{\alpha_h} \le X \le n^{\beta_h} \quad (1 \le h \le H-1), \\ R_h, & n^{\beta_h} < X < n^{\alpha_{h-1}} \quad (2 \le h \le H-1), \\ R_1, & n^{\beta_1} < X < n, \\ K, & X = n. \end{cases}$$

This function is monotone nondecreasing: it is nondecreasing on each genuine block because $A_{n,h}$ is increasing, it is constant on each gap, and the relations $R_h \le L_{h-1}$ guarantee monotonicity across adjacent pieces.

We next prove that $C_{H,n}(X) \le S_K(X)$ for all $2 \le X \le n$. If $X \in [n^{\alpha_h}, n^{\beta_h}]$, then

$$C_{H,n}(X) = \lfloor A_{n,h}(\log_n X) \rfloor \le S_K(X) \qquad \text{by the definition of } A_{n,h}.$$

If $X \in (n^{\beta_h}, n^{\alpha_{h-1}})$ with $2 \le h \le H-1$, then

$$C_{H,n}(X) = R_h \le S_K(n^{\beta_h}) \le S_K(X),$$

since S_K is nondecreasing. The same argument gives $C_{H,n}(X) = R_1 \le S_K(X)$ on (n^{β_1}, n), while $C_{H,n}(X) = 0 \le S_K(X)$ below $n^{\alpha_{H-1}}$ and $C_{H,n}(n) = K = S_K(n)$. This proves part (1).

Define

$$b_j := \inf\{X \in [2, n] : C_{H,n}(X) \ge j\} \qquad (1 \le j \le K).$$

Because $C_{H,n}$ is nondecreasing, the sequence $(b_j)_{j=1}^K$ is nondecreasing. Moreover,

$$S_K(b_j) \ge C_{H,n}(b_j) \ge j,$$

so the jth prime in the Round 15 prefix satisfies $q_j \le b_j$. This proves part (2).

For the moment asymptotics, write

$$u_j := \log_n b_j, \qquad w_j := \frac{1}{b_j} = n^{-u_j}, \qquad \mu_{H,n} := \sum_{j=1}^{K} w_j \, \delta_{u_j}.$$

Split $\mu_{H,n} = \nu_{H,n} + \lambda_{H,n}$, where $\nu_{H,n}$ is the contribution from the genuine interval blocks $L_h < j \le R_h$, and $\lambda_{H,n}$ is the contribution from the flat filler blocks.

The filler blocks have negligible reciprocal mass. For the bottom block,

$$|\lambda_{H,n}^{\text{bot}}| = L_{H-1} n^{-\alpha_{H-1}} \le A_{n,H-1}(\alpha_{H-1}) n^{-\alpha_{H-1}} = \frac{1 - \xi_H(n)}{H \alpha_{H-1} \log n} = O_H\left(\frac{1}{\log n}\right).$$

For a gap block between I_h and I_{h-1}, where $2 \le h \le H-1$,

$$|\lambda_{H,n}^{(h)}| = (L_{h-1} - R_h) n^{-\alpha_{h-1}} \le L_{h-1} n^{-\alpha_{h-1}} \le \frac{1 - \xi_H(n)}{h \, \alpha_{h-1} \log n} = O_H\left(\frac{1}{\log n}\right).$$

For the top block,

$$|\lambda_{H,n}^{\text{top}}| = (K - R_1) n^{-1} \le \frac{K}{n} = O\left(\frac{1}{\log n}\right).$$

Since there are only $O_H(1)$ filler blocks,

$$|\lambda_{H,n}| = O_H\left(\frac{1}{\log n}\right) = o_H(1).$$

We next analyze the genuine blocks. Fix $1 \le h \le H-1$. On the block $L_h < j \le R_h$, the value $u_j =: u_{h,j} \in I_h$ is defined by $A_{n,h}(u_{h,j}) = j$. Therefore, for every $f \in C^1([0,1])$,

$$\int f\, d\nu_{H,n}^{(h)} = \sum_{j=L_h+1}^{R_h} f(u_{h,j}) n^{-u_{h,j}} = \int_{\alpha_h}^{\beta_h} f(u) n^{-u}\, d\lfloor A_{n,h}(u)\rfloor.$$

Let

$$R_{n,h}(u) := \lfloor A_{n,h}(u)\rfloor - A_{n,h}(u), \qquad |R_{n,h}(u)| \le 1.$$

Applying Stieltjes integration by parts with $g(u) := f(u)n^{-u}$ gives

$$\int_{\alpha_h}^{\beta_h} g(u)\, d\lfloor A_{n,h}(u)\rfloor = \int_{\alpha_h}^{\beta_h} g(u)\, dA_{n,h}(u) + E_{n,h}(f),$$

where

$$|E_{n,h}(f)| \le 2\|g\|_\infty + \int_{\alpha_h}^{\beta_h} |g'(u)|\, du.$$

Since

$$g'(u) = \big(f'(u) - (\log n) f(u)\big) n^{-u},$$

we obtain

$$|E_{n,h}(f)| \le 2\|f\|_\infty n^{-\alpha_h} + (\beta_h - \alpha_h)\|f'\|_\infty n^{-\alpha_h} + (\beta_h - \alpha_h)(\log n)\|f\|_\infty n^{-\alpha_h} = o_H(1).$$

On the other hand,

$$n^{-u} A'_{n,h}(u) = \big(1 - \xi_H(n)\big)\rho(u)\left(1 - \frac{1}{u \log n}\right) \qquad (u \in I_h),$$

because $\rho(u) = 1/((h+1)u)$ on I_h. Hence

$$\int_{\alpha_h}^{\beta_h} f(u) n^{-u}\, dA_{n,h}(u) = \big(1 - \xi_H(n)\big) \int_{\alpha_h}^{\beta_h} f(u)\rho(u)\left(1 - \frac{1}{u \log n}\right) du = \int_{I_h} f(u)\rho(u)\, du + o_H(1),$$

uniformly for fixed f. Combining the last two displays yields

$$\int f\, d\nu_{H,n}^{(h)} = \int_{I_h} f(u)\rho(u)\, du + o_H(1).$$

Summing over h and using $|\lambda_{H,n}| = o_H(1)$, we find that for every $f \in C^1([0,1])$,

$$\int f\, d\mu_{H,n} = \int_{G_H} f(u)\rho(u)\, du + o_H(1).$$

In particular, $\mu_{H,n}$ converges weakly to the absolutely continuous measure

$$\mu_H := \rho(u)\mathbf{1}_{G_H}(u)\, du.$$

Fix $r \in \{1,2,3,4\}$, and write

$$\Sigma_r := \{(u_1, \dots, u_r) \in [0,1]^r : u_1 + \cdots + u_r \le 1\}.$$

Then

$$\int_{\Sigma_r} d\mu_{H,n}^{\otimes r} = \sum_{j_1,\dots,j_r} \mathbf{1}_{\{u_{j_1} + \cdots + u_{j_r} \le 1\}} \prod_{t=1}^{r} w_{j_t}.$$

The contribution of distinct ordered r-tuples is exactly $r!\,T_r^{(b)}(n)$, because $u_{j_1}+\cdots+u_{j_r}\le 1$ is equivalent to $b_{j_1}\cdots b_{j_r}\le n$. The repeated-index contribution is negligible:

$$\max_j w_j \le n^{-\alpha_{H-1}} \to 0,$$

while $\mu_{H,n}([0,1]) = O_H(1)$ by the weak convergence with $f\equiv 1$. Therefore

$$0 \le \int_{\Sigma_r} d\mu_{H,n}^{\otimes r} - r!\,T_r^{(b)}(n) \le \binom{r}{2}\big(\max_j w_j\big)\mu_{H,n}([0,1])^{r-1} = o_H(1).$$

Since μ_H is absolutely continuous, its product measure gives zero mass to the boundary $\partial\Sigma_r$. Weak convergence therefore implies

$$\int_{\Sigma_r} d\mu_{H,n}^{\otimes r} \to \int_{\Sigma_r} d\mu_H^{\otimes r} = \int_{\substack{u_1+\cdots+u_r\le 1\\ u_i\in G_H}} \prod_{i=1}^r \rho(u_i)\,du_1\cdots du_r = r!\,J_r^{(H)}.$$

Combining the last two displays proves part (3):

$$T_r^{(b)}(n) = J_r^{(H)} + o_H(1).$$

It remains to control the excised breakpoint region. Since $0<\rho(u)<1$ on $(0,1]$, we have

$$0 \le J_r - J_r^{(H)} \le \frac{1}{r!}\,\lambda_r\big(\{(u_1,\dots,u_r)\in[0,1]^r : \exists i,\ u_i\notin G_H\}\big),$$

where λ_r denotes r-dimensional Lebesgue measure. Moreover,

$$|(0,1]\setminus G_H| = \frac{1}{H} + 2(H-1)\tau_H \le \frac{1}{H} + \frac{1}{4H} = \frac{5}{4H}.$$

By the union bound in $[0,1]^r$,

$$\lambda_r\big(\{(u_1,\dots,u_r)\in[0,1]^r : \exists i,\ u_i\notin G_H\}\big) \le r\,|(0,1]\setminus G_H| \le \frac{5r}{4H}.$$

This proves part (4), and the final statement follows immediately by first letting $n\to\infty$ for fixed H, and then letting $H\to\infty$.

To obtain a single comparison sequence for each n, choose N_H so large that for all $n\ge N_H$ the above construction is valid and, for every $1\le r\le 4$,

$$\big|T_r^{(b)}(n) - J_r^{(H)}\big| \le \frac{1}{H}.$$

After enlarging if necessary, we may assume N_H is increasing in H. Define

$$H_*(n) := \max\{H\ge 2 : N_H\le n\},$$

and let $b_1^{(n)}\le\cdots\le b_K^{(n)}$ be the comparison sequence produced with parameter $H=H_*(n)$. Then $H_*(n)\to\infty$, and for every $1\le r\le 4$,

$$\big|T_r^{(b^{(n)})}(n) - J_r\big| \le \big|T_r^{(b^{(n)})}(n) - J_r^{(H_*(n))}\big| + \big|J_r^{(H_*(n))} - J_r\big| \le \frac{1}{H_*(n)} + \frac{5r}{4H_*(n)r!},$$

which tends to 0. This proves the final statement. □

6.4. **The prime-sequence Bonferroni comparison theorem.** The comparison theorem must be stated for *prime sequences*, not for arbitrary reals; the correct statement is the following.

Theorem 6.3 (Prime-sequence Bonferroni–4 comparison)**.** *Let $q_1 < \cdots < q_K$ be the actual odd primes played by Shortener during the Round 15 prefix, and let $p_1 < \cdots < p_K$ be any increasing sequence of odd primes with*

$$q_j \le p_j \qquad (1 \le j \le K).$$

Assume that for $1 \le r \le 4$,

$$T_r^{(p)}(n) = \Lambda_r + o(1).$$

Then

$$L(n) \le \frac{n}{2}\left(1 - \Lambda_1 + \Lambda_2 - \Lambda_3 + \Lambda_4 + o(1)\right).$$

Proof. Let A be the final played set. If the game ends before Shortener completes the prefix of length K, then $L(n) = o(n)$ and there is nothing to prove, so assume $q_1, \ldots, q_K$ were all played.

Define the odd-part map

$$\varphi(x) := \frac{x}{2^{v_2(x)}}.$$

If $\varphi(x) = \varphi(y) = m$, then $x = 2^a m$ and $y = 2^b m$. If $a \ne b$, one of x and y divides the other, contradicting that A is an antichain. So φ is injective on A.

Since each $q_j \in A$, no other element of A can be divisible by any q_j. Because q_j is odd, divisibility by q_j is preserved under the odd-part map. Hence

$$\varphi(A) \subseteq \{m \le n : m \text{ odd and } q_j \nmid m \text{ for all } j\} \cup \{q_1, \ldots, q_K\}.$$

Therefore

$$L(n) \le N(q_1, \ldots, q_K) + K = N(q_1, \ldots, q_K) + o(n),$$

where $N(q_1, \ldots, q_K)$ counts odd integers up to n avoiding all q_j.

Now replace the actual sequence q_j by the larger prime sequence p_j. The count of odd integers avoiding a finite prime set is monotone in each prime, so

$$N(q_1, \ldots, q_K) \le N(p_1, \ldots, p_K).$$

Applying fourth-order Bonferroni to the divisibility events $p_j \mid m$ over odd integers $m \le n$ gives

$$\begin{aligned} N(p_1, \ldots, p_K) \le \frac{n}{2}\Big(1 - T_1^{(p)}(n) + T_2^{(p)}(n) - T_3^{(p)}(n) + T_4^{(p)}(n)\Big) \\ + O(M_2 + M_3 + M_4), \end{aligned}$$

where M_r is the number of odd integers $\le n$ with at least r distinct prime divisors from the set $\{p_j\}$. The standard squarefree divisor-count bound gives

$$M_r = O_r\left(\frac{n(\log\log n)^{r-1}}{\log n}\right) = o(n) \qquad (2 \le r \le 4),$$

so the error is $o(n)$. Substituting $T_r^{(p)}(n) = \Lambda_r + o(1)$ yields the stated inequality. □

6.5. **The prime-rounding bridge.** It remains to convert the real comparison sequence b_j from Proposition 6.2 into an actual increasing odd-prime sequence. The naive claim

$$p_j/b_j = 1 + O(1/\log b_j)$$

uniformly in j is *not* justified by the prime number theorem alone and must not be used. The correct bridge is the half-density construction of Theorem 6.4.

Theorem 6.4 (Prime-rounding bridge)**.** *Fix $0 < \varepsilon < 1$, and let*

$$K = \left\lfloor \frac{(1-\varepsilon)n}{2\log n} \right\rfloor.$$

There exists a choice of comparison sequences

$$b_1^{(n)} \le \cdots \le b_K^{(n)}$$

from the construction of Proposition 6.2, and there exists an increasing sequence of odd primes

$$p_1 < \cdots < p_K$$

such that

$$q_j \le p_j \qquad (1 \le j \le K)$$

and

$$T_r^{(b^{(n)})}(n) = J_r + o(1) \qquad (1 \le r \le 4).$$

Moreover the primes may be chosen so that, for each fixed $1 \le r \le 4$,

$$T_r^{(p)}(n) = T_r^{(b^{(n)})}(n) + o(1) = J_r + o(1).$$

Proof. We first prove a fixed-parameter statement. Fix $H \ge 2$ and $\lambda > 1$. Let $\tau_H, \alpha_h, \beta_h, I_h, G_H$ be as in Proposition 6.2, and let $b_1, \dots, b_K$ be the sequence obtained by inverting the envelope $C_{H,n}$ for this fixed H. We shall construct increasing odd primes $p_1, \dots, p_K$ such that

$$q_j \le p_j \qquad (1 \le j \le K)$$

and, for $1 \le r \le 4$,

$$T_r^{(p)}(n) = J_r^{(H)} + O_{H,r}(\lambda - 1) + o_{H,\lambda}(1). \tag{PB1}$$

Here $n \to \infty$ with H, λ, ε fixed.

Write

$$B_{H,n}(X) := \#\{j : b_j \le X\}.$$

Since $b_j = \inf\{X : C_{H,n}(X) \ge j\}$ and $C_{H,n}$ is nondecreasing and integer-valued,

$$B_{H,n}(X) = C_{H,n}(X) \qquad (2 \le X \le n).$$

We need the following interval count, which is the point at which the half-density slack enters. Fix $\Delta > 0$. For all sufficiently large n, uniformly in $1 \le h \le H-1$ and in all X for which $[X, (1+\Delta)X] \subseteq [n^{\alpha_h}, n^{\beta_h}]$,

$$B_{H,n}((1+\Delta)X) - B_{H,n}(X) \le \left(\frac{1}{2} + o_{H,\Delta}(1)\right) \frac{\Delta X}{\log X}. \tag{PB2}$$

Indeed, put $u = \log_n X$ and $\theta = \log(1+\Delta)/\log n$. On a genuine block, $B_{H,n}(Y) = \lfloor A_{n,h}(\log_n Y) \rfloor$, up to an error of at most one at each endpoint. Therefore

$$B_{H,n}((1+\Delta)X) - B_{H,n}(X) \le A_{n,h}(u+\theta) - A_{n,h}(u) + 2.$$

For $v \in [u, u+\theta]$ and $Y = n^v$,

$$\frac{d}{dY} A_{n,h}(\log_n Y) = \frac{1 - \xi_H(n)}{(h+1)\log Y}\left(1 - \frac{1}{\log Y}\right) \le \frac{1 + o_H(1)}{(h+1)\log X}.$$

Since $h+1 \ge 2$ and $X \ge n^{\alpha_{H-1}} \ge n^{1/H}$, the endpoint error 2 is $o_{H,\Delta}(\Delta X/\log X)$. This proves (PB2).

Choose $\delta = \delta(\lambda) > 0$ with

$$(1+\delta)^2 \le \lambda.$$

All uses of the prime number theorem below are uniform for $X \geq n^{1/H}$ and for fixed multiplicative intervals, because H and δ are fixed. Thus, uniformly in such X,

$$\#\{p \text{ odd prime} : (1+\delta)X \leq p < (1+\delta)^2 X\} = (1+o_{H,\delta}(1))\frac{\delta(1+\delta)X}{\log X}. \tag{PB3}$$

Combining (PB2) and (PB3), for all sufficiently large n every multiplicative bin $[X,(1+\delta)X)$ contained in a genuine block contains strictly fewer b_j than there are odd primes in the next bin $[(1+\delta)X,(1+\delta)^2X)$.

It remains to keep the bins away from the finitely many artificial boundary layers. Choose a constant $\Gamma_H > 4$, depending only on H, large enough for the reservoir count below. For each genuine block $[n^{\alpha_h}, n^{\beta_h}]$, declare exceptional all indices j for which

$$b_j \in [n^{\alpha_h}, \Gamma_H n^{\alpha_h}] \cup [\Gamma_H^{-1} n^{\beta_h}, n^{\beta_h}],$$

and also all indices lying in the flat filler blocks of the envelope. Denote the exceptional set by $\mathcal{E}$ and its complement by $\mathcal{G}$. For fixed H and large n, these boundary reservoirs are disjoint and occur in the same order as the corresponding indices, because the gaps between successive genuine blocks have multiplicative size $n^{2\tau_H}$.

The exceptional set has negligible reciprocal mass. The flat filler blocks already satisfy the estimate proved in Proposition 6.2,

$$\sum_{j \text{ flat}} \frac{1}{b_j} = O_H\left(\frac{1}{\log n}\right).$$

For a genuine boundary layer, (PB2) with fixed multiplicative length implies

$$\sum_{\substack{j: b_j \in [X, \Gamma_H X] \\ j \text{ genuine}}} \frac{1}{b_j} \ll_{H,\Gamma_H} \frac{1}{\log n},$$

uniformly for $X \geq n^{1/H}$. There are $O_H(1)$ such layers, hence

$$\sum_{j \in \mathcal{E}} \frac{1}{b_j} = O_{H,\lambda}\left(\frac{1}{\log n}\right). \tag{PB4}$$

We next assign primes to the exceptional indices. On each boundary reservoir of the form $[X, \Gamma_H X]$, the number of genuine exceptional indices is at most

$$\left(\frac{1}{2} + o_H(1)\right)\frac{(\Gamma_H - 1)X}{\log X},$$

by (PB2) applied with $\Delta = \Gamma_H - 1$, while the number of odd primes in the same reservoir is

$$(1+o_H(1))\frac{(\Gamma_H - 1)X}{\log X}.$$

At a left endpoint one must also place the adjacent flat atom. Its size is at most the value of the envelope at that endpoint, namely

$$O_H\left(\frac{X}{\log X}\right),$$

with an implied constant depending only on H; the choice of Γ_H ensures that the reservoir contains enough odd primes for both the flat atom and the genuine boundary layer. More explicitly, the sorted lower bounds b_j in the reservoir satisfy the one-dimensional Hall condition: the full reservoir has enough primes by the preceding choice of Γ_H, and every proper suffix $[Y, \Gamma_H X]$ contains at most half as many genuine exceptional b_j as there are odd primes there, by the interval estimate (PB2), uniformly for multiplicative lengths at most Γ_H, and the prime number theorem. Thus greedy assignment to the smallest unused odd prime which is at least the current b_j succeeds inside

the reservoir. The top flat block is placed in $[n, \Gamma_H n]$, which has enough odd primes because $K \le n/(2\log n)$ and $\Gamma_H > 2$. Consequently all exceptional indices can be assigned, in increasing order, to distinct odd primes satisfying $b_j \le p_j$ and hence $q_j \le p_j$. Their total reciprocal mass is

$$\sum_{j\in\mathcal{E}} \frac{1}{p_j} = O_{H,\lambda}\left(\frac{1}{\log n}\right). \tag{PB5}$$

Now consider the nonexceptional indices. In each genuine block, partition the remaining interval into consecutive multiplicative bins

$$[X, (1+\delta)X),$$

discarding at most the first and last partial bin into $\mathcal{E}$. For each full bin, assign the b_j in that bin, in increasing order, to the odd primes in the next bin. The count comparison (PB2)-(PB3) shows that this is possible for all large n. The prime bins are disjoint and increasing, so together with the reservoir assignments this produces one increasing sequence of odd primes. For every $j \in \mathcal{G}$,

$$b_j \le p_j \le (1+\delta)^2 b_j \le \lambda b_j. \tag{PB6}$$

We compare moments. The total reciprocal mass of the b_j is $O_H(1)$ by Proposition 6.2; the same is true for the p_j, by (PB5) and (PB6). Hence every r-tuple containing at least one exceptional index contributes

$$O_{H,r,\lambda}\left(\frac{1}{\log n}\right) = o_{H,\lambda}(1)$$

to either $T_r^{(b)}(n)$ or $T_r^{(p)}(n)$. It remains to compare tuples lying wholly in $\mathcal{G}$.

For such a tuple $I = \{j_1 < \cdots < j_r\}$, set

$$b_I := \prod_{j\in I} b_j, \qquad p_I := \prod_{j\in I} p_j.$$

By (PB6),

$$b_I \le p_I \le \lambda^r b_I, \qquad \lambda^{-r}\frac{1}{b_I} \le \frac{1}{p_I} \le \frac{1}{b_I}.$$

The weight change contributes $O_{H,r}(\lambda - 1)$, because the total b-reciprocal mass is $O_H(1)$. The only remaining discrepancy is the product boundary

$$n/\lambda^r < b_I \le n.$$

For fixed H and λ, this boundary layer is $o_{H,\lambda}(1)$: in logarithmic coordinates it is a strip of width $r\log\lambda/\log n$ around the hyperplane $u_1 + \cdots + u_r = 1$, and the empirical measures from Proposition 6.2 converge to an absolutely continuous limit. Equivalently, one may partition into unit logarithmic cells $[e^a, e^{a+1})$; each cell has reciprocal mass $O_H(1/\log n)$ by (PB2), and only $O_{H,\lambda}((\log n)^{r-1})$ cell patterns meet the strip. Thus

$$T_r^{(p)}(n) = T_r^{(b)}(n) + O_{H,r}(\lambda - 1) + o_{H,\lambda}(1).$$

Using Proposition 6.2, $T_r^{(b)}(n) = J_r^{(H)} + o_H(1)$, and this proves (PB1).

It remains to diagonalize. Choose a sequence $H_m \to \infty$ such that

$$|J_r^{(H_m)} - J_r| \le \frac{1}{m} \qquad (1 \le r \le 4).$$

For each m, choose $\lambda_m > 1$ so close to 1 that the constants implicit in $O_{H_m,r}(\lambda_m - 1)$ in (PB1) are at most $1/m$ for all $1 \le r \le 4$. For each m, the fixed-parameter construction above is valid for all $n \ge N_m$ after enlarging N_m so that also

$$|T_r^{(b)}(n) - J_r^{(H_m)}| \le \frac{1}{m},$$

for every $1 \le r \le 4$. We may take N_m increasing. For $N_m \le n < N_{m+1}$ use the construction with $H = H_m$ and $\lambda = \lambda_m$. This gives $H(n) \to \infty$ slowly enough that the smallest genuine scale

$n^{\alpha_{H(n)-1}}$ still tends to infinity, so the uniform prime-number estimates used above remain valid. The resulting comparison and prime sequences satisfy, for every fixed $1 \le r \le 4$,

$$T_r^{(b^{(n)})}(n) = J_r + o(1), \qquad T_r^{(p)}(n) = T_r^{(b^{(n)})}(n) + o(1) = J_r + o(1).$$

□

6.6. **The finite Bonferroni–4 theorem.**

Theorem 6.5. *Under the Round 15 strategy σ_{15},*

$$L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n.$$

Proof. By Theorem 6.4, there are a comparison sequence $b_1 \le \cdots \le b_K$ and an increasing odd-prime sequence p_j with $q_j \le p_j$ such that

$$T_r^{(b)}(n) = J_r + o(1) \qquad (1 \le r \le 4).$$

The same theorem gives

$$T_r^{(p)}(n) = J_r + o(1) \qquad (1 \le r \le 4).$$

Applying Theorem 6.3 with $\Lambda_r = J_r$ gives

$$L(n) \le \frac{n}{2}\left(1 - J_1 + J_2 - J_3 + J_4 + o(1)\right) = \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n.$$

Numerical evaluation yields

$$\frac{\mathcal{W}_4}{2} = 0.1897112 < 0.19,$$

so the theorem follows. □

Remark 6.6. The Lean file `Round15Bonferroni4/Target.lean` formalizes the endgame reduction: once the first four moment terms are componentwise within 10^{-5} of the certified constants $J_1, \ldots, J_4$, the strict inequality $L(n) < 0.19n$ follows eventually. Theorems 6.3 and 6.4 are the rigorous prose ingredients that supply those moment hypotheses.

7. Restricted Carrier Classes

This section records two positive theorems showing that several natural Prolonger classes are not genuinely hard. In both cases Shortener resolves all small-prime structure after only $O_\alpha(n/\log n)$ moves, leaving at most $O_\alpha(n/\log n)$ total legal play.

7.1. **Disjoint small-prime carriers.**

Theorem 7.1 (Theorem 5)**.** *Fix α with $1/3 < \alpha < 1/2$, and set $y = n^\alpha$. Suppose every composite Prolonger move has all prime factors at most y, and suppose further that distinct composite Prolonger moves have pairwise-disjoint prime supports. Then Shortener has an online strategy forcing*

$$L(n) = O_\alpha\left(\frac{n}{\log n}\right).$$

Proof sketch. Let B be the set of small primes that occur in some composite Prolonger move. For each $p \in B$, let $C(p)$ denote its unique carrier and let $e(p) = v_p(C(p))$.

Shortener uses a three-phase priority strategy:

(1) whenever a legal prime exists, play a legal prime;
(2) if $C(p)$ has at least two distinct prime factors, earmark the repair move $p^{e(p)+1}$ and play it whenever legal and higher-priority moves are unavailable;
(3) for distinct carrier primes $p, q \in B$ with $C(p) \ne C(q)$, earmark the semiprime pq and play it whenever legal and higher-priority moves are unavailable.

The disjoint-support hypothesis guarantees legality: a power repair $p^{e(p)+1}$ is incomparable with its own carrier because that carrier contains another prime, and incomparable with every other carrier because no other carrier uses p; similarly, a cross-carrier pair pq is incomparable with all earlier phase-2 powers and all composite carriers.

Once these repairs are exhausted, every prime $p \le y$ is resolved. Indeed, if some legal move x were still divisible by p, then every prime factor of x would lie in B, and either x would involve primes from two distinct carriers, in which case a phase-3 pair would divide x, or all prime factors of x would lie inside one carrier, in which case either x is comparable with that carrier or some phase-2 power repair divides x. Both cases are impossible.

Once every $p \le y$ is resolved, every future legal move lies in the sparse set

$$\mathcal{S}_y := \{x \le n : P^-(x) > y,\ \Omega(x) \le 2\},$$

whose size is $O_\alpha(n/\log n)$. The total move count is therefore $O_\alpha(n/\log n)$. □

7.2. Squarefree rank-≤ 3 carriers.

Theorem 7.2 (Theorem 6)**.** *Fix α with $1/3 < \alpha < 1/2$, and set $y = n^\alpha$. Suppose every composite Prolonger move is squarefree, supported on primes at most y, and of rank at most 3. Then Shortener has an online strategy forcing*

$$L(n) = O_\alpha\left(\frac{n}{\log n}\right).$$

Proof sketch. Let B be the set of carrier primes. Shortener again uses a priority strategy:

(1) legal primes;
(2) squares p^2 for primes $p \in B$ not already played;
(3) squarefree semiprimes pq on B;
(4) squarefree triples pqr on B.

After phases 1 and 2, every legal survivor is squarefree and supported on B. Indeed, any prime divisor outside B would already have been resolved by a phase-1 prime move, and any repeated prime factor p^2 would have been removed by a phase-2 square. From there the phase-3 and phase-4 priorities eliminate all rank-2 and rank-3 squarefree survivors directly.

The only nontrivial case is a surviving squarefree integer x with at least four prime factors from B. Choose three of them, say p, q, r. Then $pqr \mid x$. If the triple pqr was legal after phase 3, phase 4 would have played it. If it was not legal, then some earlier move comparable with pqr already existed. Because all carriers are squarefree of rank at most 3, any such earlier comparable move must itself divide x; in every case x is therefore illegal. So no legal move survives the four phases.

Finally, the number of played phase-1, phase-2, and phase-3 moves is $O_\alpha(n/\log n)$ by trivial prime and semiprime counts, while the phase-4 triple count is also $O_\alpha(n/\log n)$ for each fixed $\alpha < 1/2$. Hence the total game length is $O_\alpha(n/\log n)$. □

Remark 7.3. These theorems explain why the later universal carrier-mass obstruction must be read correctly. The block-product family is an obstruction to a *proof method*, not to the actual existence of strong Shortener strategies. In fact, block-product play lies inside the scope of Theorem 7.1.

8. Proof-Class Barrier Taxonomy

This section records rigorous results showing that several natural proof programs cannot close the sharp-rate gap. Unlike the structural negatives of Section 9, the emphasis here is methodological: each subsection isolates a named class of arguments and proves that the class is insufficient. We also keep the strategy-dependence labels explicit. In particular, the headline upper bound of Theorem 6.5 is proved for the specific Shortener strategy σ_{15} and does not use the max-unresolved-harmonic-degree strategy σ^*.

8.1. **Transversal-Integrality and Sherali–Adams.** The first barrier is purely static. It shows that low-level fractional or Sherali–Adams relaxations cannot certify the large transversals needed for the late separator program.

Theorem 8.1 (Transversal-integrality barrier)**.** *Fix $0 < \alpha < 1$. Let P be an N-point set, put*

$$\ell = \lfloor \log N \rfloor, \qquad q = \lfloor \alpha N \rfloor,$$

and write $\tau_{\mathbb{Z}}(\mathcal{C})$ and $\tau_f(\mathcal{C})$ for the integral and fractional transversal numbers of an ℓ-uniform family $\mathcal{C} \subseteq \binom{P}{\ell}$. Then for all sufficiently large N there exists such a family $\mathcal{C}$ with

$$\tau_{\mathbb{Z}}(\mathcal{C}) > \alpha N, \qquad \tau_f(\mathcal{C}) \le \frac{N}{\ell}.$$

Moreover, for every ℓ-uniform family $\mathcal{H} \subseteq \binom{P}{\ell}$ and every $0 \le r < \ell$,

$$\mathrm{SA}_r(\mathcal{H}) \le \frac{N}{\ell - r}.$$

In particular, if $r \le \ell/2$, then

$$\mathrm{SA}_r(\mathcal{C}) \le \frac{2N}{\ell} = (2 + o(1))\frac{N}{\log N},$$

while $\tau_{\mathbb{Z}}(\mathcal{C}) > \alpha N$, so the level-$r$ integrality gap is at least $(\alpha/2 + o(1)) \log N$.

Proof sketch. The family $\mathcal{C}$ is a random q-avoiding covering design. One chooses

$$s \asymp \delta^{-1} \log \binom{N}{q}, \qquad \delta = \frac{\binom{N-q}{\ell}}{\binom{N}{\ell}},$$

independent ℓ-subsets of P. The union bound shows that, with positive probability, every q-subset of P misses at least one chosen block; this is equivalent to $\tau_{\mathbb{Z}}(\mathcal{C}) > q \ge \alpha N + O(1)$. On the other hand, the uniform fractional weight ℓ/N on each vertex gives $\tau_f(\mathcal{C}) \le N/\ell$.

The Sherali–Adams estimate is cleaner and fully deterministic. For an ℓ-uniform family $\mathcal{H}$, the standard symmetric level-r assignment with marginal $(\ell - r)/N$ is feasible, yielding $\mathrm{SA}_r(\mathcal{H}) \le N/(\ell - r)$. The resulting gap is therefore already logarithmic at all levels $r \le \ell/2$. □

Remark 8.2. This theorem is strategy-independent: it is a static obstruction to any proof that hopes to derive a strong game upper bound from low-level separator LPs or their Sherali–Adams tightenings.

8.2. **Fixed-Rank q-Shadow / Covering Dichotomy.** The second barrier isolates the precise point where the spectral packet program can fail. Either there is an honest legal separator of the expected size, or Shortener's own previously created q-shadow already occupies almost all of the separator layer.

Theorem 8.3 (Fixed-rank q-shadow dichotomy)**.** *Let P be a set of size $K = h + L$ with $h, L \ge 1$, and fix $1 \le q \le h$. Set*

$$Y = \binom{P}{L}, \qquad Q_q = \binom{P}{q}.$$

For a blocker family $\mathcal{D} \subseteq 2^P$ and a played complement family $\mathcal{C} \subseteq Y$, let $F_q(\mathcal{D}) \subseteq Q_q$ be the comparability shadow of $\mathcal{D}$ on the q-layer, let $B_q(C) = \{Q \in Q_q : Q \cap C = \varnothing\}$, and define the legal separator set

$$A_q(\mathcal{D}, \mathcal{C}) := Q_q \setminus \Big(F_q(\mathcal{D}) \cup \bigcup_{C \in \mathcal{C}} B_q(C) \Big).$$

If $R \subseteq Y$ has density

$$r = \frac{|R|}{|Y|} > 0$$

and no legal q-separator in $A_q(\mathcal{D},\mathcal{C})$ captures at least $\frac{1}{2}\delta_q|R|$ live complements, where

$$\delta_q = \frac{\binom{K-q}{L}}{\binom{K}{L}}, \qquad \lambda_q^2 = \frac{h(K-q)}{qL},$$

then

$$\sigma_q(\mathcal{D}) + |\mathcal{C}|\delta_q > 1 - \frac{4\lambda_q^2}{r}, \qquad \sigma_q(\mathcal{D}) := \frac{|F_q(\mathcal{D})|}{|Q_q|}.$$

At the central scale $q \sim 2(\log h)^2$, $L \sim h/\log h$, one has

$$\delta_q = (e + o(1))h^{-2}, \qquad \lambda_q^2 = (2 + o(1))\frac{\log h}{h},$$

so if $\sigma_q(\mathcal{D}) = o(1)$, then necessarily

$$|\mathcal{C}| > \left(\frac{1}{e} - o(1)\right) h^2.$$

Proof sketch. The proof packages the Johnson-scheme separator theorem in the exact form needed for the packet game. If no legal q-separator captures half of the expected mass, then the live set R has negligible projection onto the legal separator subspace; quantitatively this forces $|A_q(\mathcal{D},\mathcal{C})|/|Q_q| < 4\lambda_q^2/r$. Rewriting the complement of A_q gives the displayed bound on $\sigma_q(\mathcal{D}) + |\mathcal{C}|\delta_q$.

The central-scale constants are then obtained by direct asymptotics for δ_q and λ_q^2. The upshot is conceptual: spectral starvation is possible only after Shortener has already accumulated a macroscopic amount of her own q-shadow or has played on the order of h^2 complements. □

8.3. **The Separator-Only Barrier.** We next record the cleanest theorem to emerge from the late separator-first program: any proof that uses only finitely many odd lower-half separators, with no genuine composite fallback, cannot yield even an $o(n)$ upper bound.

Theorem 8.4 (Finite odd-carrier separator-only barrier)**.** *Let $r_1(n) = n(\log\log n)^2/\log n$. No finite odd-carrier separator-only closure can prove $L(n) = O(r_1(n))$. More precisely, if $S_n \subseteq [2, n/2]$ is an odd separator family with*

$$|S_n| = o(r_1(n)), \qquad \sum_{p|s \text{ for some } s\in S_n} \frac{1}{p} = o(1),$$

and if the only lower-half composite moves used by the method are elements of S_n, then Prolonger has a legal prefix of length $o(r_1(n))$ after which all separators in S_n are dead and at least

$$\frac{n}{2} - o(n)$$

elements of U remain legal.

Proof sketch. If $s \in S_n$ is still legal, Prolonger plays the dyadic multiple $x = 2^a s \in (n/2, n]$, choosing a so that $n/2 < x \le n$. Since all elements of S_n are odd, any previous lower-half separator comparable to x would have been comparable to s already, contradicting the legality of s. Distinct upper-half moves are automatically incomparable. Thus each legal separator can be preempted at unit cost.

After at most $2|S_n| = o(r_1(n))$ turns, every separator is dead. The remaining illegal elements of U are those already played in the upper half, together with the multiples of the prime carrier set of S_n; the reciprocal carrier mass assumption bounds the latter by $o(n)$ via the union bound. Hence $|U_{\text{legal}}| = n/2 - o(n)$ at the end of the prefix. □

Remark 8.5. The theorem is a class-level statement. It does not say that every separator idea fails; it says that a proof using only finitely many odd separators and no composite fallback cannot close the game. This is the correct reading of the failure of the separator-first route and of the specific strategy τ_{SF}.

8.4. **Strategy-Dependence Classification.** The previous subsections are all strategy-independent. The next result is a meta-theorem describing which later ingredients survive strategy replacement and which do not.

Proposition 8.6 (Strategy-dependence classification)**.** *The rigorous ingredients of the program partition into three classes.*

(1) Strategy-independent *results: Shield Reduction, Theorem A, T2, the Round 15 σ_{15} upper bound, the separator theorems of Theorems 8.1, 8.3 and 8.4, and the structural negatives of Section 9.*
(2) σ^*-dependent *results: the R35 state inequality, online harmonic domination, and the smallest-legal-prime rule in its unrestricted form.*
(3) σ^*-specific *results: the cooperative-embedding construction defeating σ^* at cost $n^{o(1)}$, the ST-capture refutation in its forced-leaf formulation, and the residual-floor diagnosis attached to the σ^* bookkeeping framework.*

In particular, the load-bearing ingredients of the older packet closure program do not survive strategy replacement, whereas the headline upper bound of Theorem 6.5 is independent of σ^*.

Discussion. This is a classification theorem rather than a new counting argument. The R54 audit tracks, item by item, which proofs use only legality and combinatorics, which rely on the max-unresolved-harmonic-degree rule σ^*, and which are refutations aimed specifically at the states generated by σ^*. The cooperative-embedding theorem against σ^* is decisive here: it shows that vulnerability of σ^* cannot by itself be promoted to a minimax barrier against all Shortener strategies. □

8.5. **Conditional σ^* Results.** We record the most useful σ^*-dependent statements for completeness, but we emphasize that none of them is used in Section 6. Their role is diagnostic, not foundational.

Proposition 8.7 (Conditional σ^* tools)**.** *Within the σ^* framework one has the following rigorously derived, but conditional, statements.*

(1) *The R35 state inequality and the online harmonic-domination estimate control the cumulative useful mass only under the σ^* move rule.*
(2) *The smallest-legal-prime principle is valid only under the additional hypothesis that every legal composite has a legal prime factor.*
(3) *The ST-capture inequality is false in general, even along σ^*-reachable forcing traces.*
(4) *Even the idealized residual-floor regime with $\mu(F_{\text{useful}}) = 0$ does not by itself reach the conjectural $r_1(n)$ scale.*

References. Items (1) and the dependence on σ^* are catalogued in the R54 audit. Item (2) is the safe positive lemma isolated in the Round 46 note. Item (3) is the ST-capture refutation of Round 46, and item (4) is the residual-floor diagnosis of Round 44. Together they explain why the older state-inequality program is valuable as a diagnostic framework but cannot presently support an unconditional sharp upper bound. □

Remark 8.8. For reviewers, the key point is simple: the headline theorem of Section 6 is proved for σ_{15} and does not use any of the conditional material in this subsection.

9. Structural Negatives

We collect here a second family of rigorous no-go results. Unlike the proof class barriers of Section 8, these statements are best read as structural diagnoses of particular candidate mechanisms. Each one says that a natural route fails for a concrete mathematical reason.

9.1. **Universal Block-Product Carrier-Mass Counter.**

Proposition 9.1. *For every Shortener strategy there exists a Prolonger response P such that*

$$\sum_{p\in B(P)} \frac{1}{p} \geq \frac{1}{2}\log\log n + O(1),$$

where $B(P)$ is the set of prime carriers used by Prolonger's moves. Consequently no proof of $L(n) = O(n/\log n)$ can proceed by first showing that a Shortener strategy forces uniformly bounded reciprocal carrier mass.

Proof sketch. Partition the primes $p \leq n^{1/3}$ into disjoint blocks whose products lie in $(\sqrt{n}, n^{5/6}]$. These block products are pairwise coprime, so a single Shortener move can kill at most one of them. Greedy descending-by-mass play then lets Prolonger secure at least half of the total reciprocal mass of the small primes, namely $(1/2)\log\log n + O(1)$. The standard smooth-number sieve would convert a uniform $O(1)$ carrier-mass bound into $O(n/\log n)$, so the carrier-mass route is dead in that form. □

9.2. **Separate-Rank Fan No-Go.**

Proposition 9.2. *Let $H(n) = \sum_{p\leq n^\delta} 1/p \sim \log\log n$. In the fixed-rank upper-half fan architecture,*

$$W_h(n) \asymp \frac{n}{\log n}\frac{H(n)^h}{h!},$$

so

$$\max_h W_h(n) \asymp \frac{n}{\sqrt{\log\log n}} = o(n).$$

With the proved divisor-shadow retention factor $2^{-2(2^h-1)}$, the total separate-rank contribution satisfies

$$\sum_{h\geq 1} F_h(n) = \frac{n}{(\log n)^{1-o(1)}} = o(n).$$

Hence neither a single rank nor a rank-by-rank summation can yield a linear lower bound.

Proof sketch. The single-rank ceiling follows by optimizing $H^h/h!$ at $h \asymp H$ and applying Stirling. For the full route, one maximizes

$$F_h(n) = \frac{n}{\log n}\frac{H^h}{h!}\,2^{-2(2^h-1)}$$

over h; the optimum occurs at $h \asymp \log_2\log\log\log n$, and even there the gain is only $(\log n)^{o(1)}$. Summing over all ranks changes only the $o(1)$ term in the exponent. □

9.3. **Multi-Rank Same-b Coupling Obstruction.**

Proposition 9.3. *The obvious same-b multi-rank coupling mechanism for the fan hierarchy is blocked twice:*

1. *same-b upper-half cores cannot be nested (Sperner obstruction);*
2. *each earlier compatible move shields at most one top-lateral divisor of a given upper-half target (LCM obstruction).*

Consequently the hoped-for uniform multi-rank shielding theorem cannot be derived from same-b coupling.

Proof sketch. If $S \subsetneq T$ and both $A_S b$ and $A_T b$ lie in $(n/2, n]$, then $A_T b = A_S b \cdot A_{T\setminus S} \geq 2A_S b > n$, impossible. So the same-b cores form a Sperner family rather than a nested chain. For the LCM obstruction, if one earlier move $z \leq n$ shielded two top-lateral divisors of $y = A_T b$, then the least common multiple of those divisors would equal y, forcing $y \mid z$ and hence $z = y$, contradicting the requirement that z be an earlier distinct move. □

9.4. **Residual-Width Refutation.**

Proposition 9.4. *The domination-only residual-width lemma is false. More precisely, for every certificate family $C \subseteq \{2, \dots, n\}$ there exists an antichain $A \subseteq U$ with $|A| \le |C|$ such that $C \subseteq \mathrm{Comp}(A)$ and*

$$w(R(A)) \ge \frac{n}{2} - |C| - O(1).$$

Thus residual width cannot be controlled from $|C|$ and domination data alone.

Proof sketch. For $c \in C$, let $\lambda_n(c)$ be the least multiple of c in $(n/2, n]$. Then $A = \{\lambda_n(c) : c \in C\}$ is an antichain in U, each $c \in C$ is comparable with some element of A, and every untouched element of U is incomparable with all of A. Hence $U \setminus A \subseteq R(A)$, which already gives the lower bound on the residual width. □

9.5. **Ford-Route Correction.**

Proposition 9.5. *The old Ford-band reduction based only on a factorization $u = am$ with $a \in (Y, 2Y]$ and $P^-(m) > Y^\delta$ is invalid. Roughness of the cofactor is not enough: Ford's divisor-interval theorem applies to integers that are themselves rough, and the naive bandwise conclusion can fail by a full factor of $\log n$.*

Proof sketch. The counterexample takes a dyadic band of smooth squarefree skeletons $a = 30c \in (Y, 2Y]$ with constant reciprocal mass and multiplies them by large primes $\ell \in (n/(2a), n/a]$. Every resulting upper-half integer has a factorization $u = a\ell$ with rough cofactor ℓ, but there are $\gg n/\log n$ such numbers in one band, not $O(n/\log^2 n)$. So the reduction was importing the wrong theorem; any valid Ford-based route must control the roughness of the whole survivor or add separate bandwise reciprocal-mass control. □

9.6. **Band-Local Closure Explosion.**

Proposition 9.6. *Fix $\lambda > 1$ and $0 < \gamma < 1$. Let*

$$\mathcal{P}_{\mathrm{hi}} = \{p \text{ prime} : X \le p \le X^\lambda\}, \qquad \mathcal{C}_X(\gamma) = \{c \le X^\gamma : \mu^2(c) = 1\},$$

and let $\mathcal{B}_X(\lambda)$ be the set of triple products pqr with $p < q < r$ in $\mathcal{P}_{\mathrm{hi}}$. Then some dyadic band $(Y, 2Y]$ contains a set

$$\mathcal{A}_Y \subseteq \{bc : b \in \mathcal{B}_X(\lambda),\ c \in \mathcal{C}_X(\gamma)\}$$

with

$$\sum_{a \in \mathcal{A}_Y} \frac{1}{a} \gg_{\lambda,\gamma} 1.$$

If the proper divisors of the elements of $\mathcal{A}_Y$ are already unavailable and $Y \le n^{1-\eta}$, this single band supports $\gg_{\lambda,\gamma,\eta} n/\log n$ legal upper-half moves.

Proof sketch. The reciprocal mass of the triple products in the high band is constant:

$$\sum_{p<q<r \in \mathcal{P}_{\mathrm{hi}}} \frac{1}{pqr} \gg_\lambda 1.$$

The reciprocal mass of the low squarefree pool is logarithmic:

$$\sum_{c \in \mathcal{C}_X(\gamma)} \frac{1}{c} \asymp_\gamma \log X.$$

Their product therefore has total reciprocal mass $\gg \log X$, and pigeonholing over only $O(\log X)$ dyadic bands yields one band with constant mass. The upper-half lift then follows from the prime number theorem in the intervals $(n/(2a), n/a]$. □

9.7. Directed Rank-3 Cleanup Is Not Budget-Limited.

Proposition 9.7. *For a fixed-power band $I = (X, X^\lambda] \subseteq [2, n^\alpha]$ with $1/3 < \alpha < 1/2$, let*

$$T_I(n) = \#\{p < q < r : \ p, q, r \in I, \ pqr \le n/2\}.$$

Then

$$T_I(n) \ll_{\alpha,\lambda} \frac{n}{\log n}, \qquad \sum_I T_I(n) \ll_{\alpha,\lambda} \frac{n \log\log n}{\log n}.$$

Thus the natural same-band rank-3 cleanup is not blocked by raw move count; the real obstruction is online fiber persistence after Prolonger preempts one triple.

Proof sketch. For fixed $p, q \in I$, the admissible primes r satisfy $r \le n/(2pq)$, so the prime number theorem bounds their number by $\ll n/(pq \log n)$. Summing over $p < q$ in one fixed-power band and using Mertens on that band gives $T_I(n) \ll n/\log n$. Since there are only $O(\log\log n)$ such bands up to n^α, the total budget is $O(n \log\log n/\log n)$, well below the T2 scale. □

9.8. Dyadic-Fiber Positive-Density Collapse.

Proposition 9.8. *A positive-density theorem for one dyadic fiber is equivalent in strength to a linear lower bound for the original game. More precisely, if Prolonger could force ηK moves inside one fiber*

$$\mathcal{T}_b = \{bc : K/2 < c \le K\}, \qquad b > \sqrt{n},$$

then $L(K) \gg_\eta K$ for the original divisibility game at scale K.

Proof sketch. Inside one fixed b-fiber, divisibility is exactly divisibility in the scaled game on $[2, K]$, because

$$bd_1 \mid bd_2 \iff d_1 \mid d_2.$$

Moreover, different large-prime fibers are laterally incomparable when $b > \sqrt{n}$. So a Shortener strategy can play independently inside each fiber, and a positive-density lower bound in one fiber would already imply a positive density lower bound for the original upper-half divisibility game at scale K. In other words, the would-be auxiliary theorem is just the main linear conjecture in disguise. □

10. Formal Verification

Shield Reduction (Theorem 2.2) is formally verified in Lean 4 via Aristotle with no remaining sorries. For the exact $5n/24 + O(1)$ cover theorem (Theorem 4.2), the explicit cover/packing identities are likewise formalized with no remaining sorries, while the final floor-sum cardinality count remains elementary prose. The 13/36 upper bound (Theorem 5.1) is likewise formally verified with no remaining sorries. Theorem A (Theorem 3.1) and the 5/16 upper bound (Theorem 5.3) have Lean artifacts with classical-number-theory sorries (PNT- and Chebyshev-ϑ-adjacent, enumerated in Appendix C); these are Mathlib-dependency gaps rather than logical holes. T2 has a Lean-verified finite core with the activation-stage wrapper pending. The endgame reduction for the Bonferroni–4 theorem — namely, that if the four Round 15 moment terms are componentwise within 10^{-5} of their certified limits $J_1, \ldots, J_4$ plus eventual $o(1)$ error, then $L(n) < 0.19n$ eventually — is formally verified in Lean 4 with no remaining sorries in `erdos_872_core/RequestProject/Round15Bonferroni4/Target.lean`. The supplying moment-convergence hypotheses of Theorems 6.3 and 6.4 are proved rigorously in prose in Section 6, with multi-verifier audit and sandbox computation.

All paths in Table 1 are relative to the public artifact repository [Bud26], with the Lean-specific paths rooted at `erdos-872/lean/`. Appendix C lists the current explicit `sorry` sites in the checked-in files rather than the older aggregate counts recorded in the repository index.

Result	**Lean artifact**	**Sorries**	**Status note**
Shield Reduction	`shield_reduction/shield_reduction_aristotle/RequestProject/ShieldReduction.lean`	0	Complete formal proof of the terminal inequality
Theorem A / T1	`theorem_A/theorem_A_shield_lower_bound_aristotle/RequestProject/ShieldMainTheorem.lean`	5	All remaining sorries are PNT-adjacent analytic inputs
Exact 5/24 cover	`tau_5_24/tau_5_24_aristotle/RequestProject/\{Cover,Packing,Tau\}.lean`	0	Structural identities are formalized; the floor-sum cardinality count remains elementary prose
13/36 upper bound	`shortener_13_36/shortener_13_36_v2_aristotle/RequestProject/Shortener/MainTheorem.lean`	0	Complete formal proof for the truncation-based strategy
5/16 upper bound	`shortener_5_16/shortener_5_16_aristotle/RequestProject/Shortener516/Theorems.lean`	1	One bundled Chebyshev / prefix-legality gap remains; the combinatorial core is zero-sorry
T2 finite core	`erdos_872_core/RequestProject/T2Finite/*.lean`	0 in current files	Graph/hypergraph potentials and residual comparison are formalized; the activation-stage wrapper is still being integrated
Round 15 endgame reduction	`erdos_872_core/RequestProject/Round15Bonferroni4/Target.lean`	0	Formalizes `eventually_strict_lt_point19_of_componentwise` the analytic moment convergence is supplied by Theorems 6.3 and 6.4

Table 1. Current Lean / Aristotle verification status.

Remark 10.1. Two distinctions are important. First, a zero-sorry reduction is not the same thing as a full end-to-end formalization of the surrounding theorem: the Round 15 Lean file closes the last step of the proof, while the analytic moment-convergence input remains in prose. Second, the T2 project already contains zero-sorry formalizations of the finite potential game and residual comparison layers, but not yet a single final Lean theorem packaging the full activation-stage argument of Theorem 3.4.

11. Conclusion

We have proved

$$\left(\frac{1}{8} - o(1)\right) \frac{n \log\log n}{\log n} \le L(n) \le \left(\frac{\mathcal{W}_4}{2} + o(1)\right) n = (0.1897112 + o(1))n < 0.19n,$$

and, through T2, we have also shown that the lower-bound side naturally reaches the larger scale

$$\frac{n(\log\log n)^2}{\log n}.$$

The exact order of growth therefore remains open. What has changed is the shape of the remaining problem. The present barrier inventory rules out a large number of natural separator, packet, closure, and fan-based routes, so the sharp-rate question is no longer just a matter of refining the first successful heuristic.

Three directions now appear genuinely distinct. The first is an improved upper bound, ideally an $o(n)$ strategy that goes beyond the odd-prime-prefix family of Sections 5 and 6. The second is a lower bound that exceeds the current T2 scale and survives the fan, fiber, and same-core obstructions of Section 9. The third is a clearer understanding of restricted adversary classes and

proof classes: theorems such as Theorems 7.1, 7.2, 8.1 and 8.4 suggest that the game mixes genuine combinatorial rigidity with strong sensitivity to the exact modeling framework.

The first-player convention also remains mathematically relevant. Throughout this paper Prolonger moves first, matching the forum literature and all prior private rounds in the repository. A Shortener-first version of the game could have a different asymptotic behavior and deserves separate study.

Appendix A. Full Proof of the T2 Lower Bound

This appendix proves Theorem 3.4. The proof is self-contained: all target families, auxiliary games, weights, activation rules, deletion errors, and asymptotic estimates are defined below.

Throughout this appendix n is large, $0 < \delta < 1/4$ is fixed, and

$$Y := n^{\delta}.$$

Let $\mathcal{P}_Y$ be the set of odd primes at most Y. For $a < c$ in $\mathcal{P}_Y$, put

$$I_{a,c} := \left(\frac{n}{2ac}, \frac{n}{ac}\right], \qquad B_{a,c} := I_{a,c} \cap \mathbb{P},$$

and define the target family

$$\mathcal{T} := \{\, acb : \ a < c, \ a, c \in \mathcal{P}_Y, \ b \in B_{a,c} \,\}.$$

For large n, every such b is greater than Y: indeed $b > n/(2Y^2) = \frac{1}{2}n^{1-2\delta} > Y$. Thus a, c, b are distinct primes and each target acb lies in $(n/2, n]$.

Proposition A.1 (Finite scaled capture game)**.** *Let H be a finite weighted incidence system. Its vertices form a finite set V, its edges form a finite set $\mathcal{E}$, each edge e has a non-negative weight $w(e)$, and each incidence set $e \subseteq V$ has size at most 3. Consider the following Maker-first scored game.*

At any time some vertices are captured, some vertices are deleted, and some edges are scored. An edge is live if it is not scored and is disjoint from the deleted vertices. Maker chooses a live edge f, scores $w(f)$, and captures all vertices of f. Breaker then either deletes one uncaptured vertex or scores one live edge. Let S be the total scored weight, by either player, and for a live edge e define

$$\Phi(e) := 2^{|e \cap C|} w(e),$$

where C is the current captured-vertex set. Scored and deleted edges contribute no live potential. Set

$$Q := 8S + \sum_{e \text{ live}} \Phi(e).$$

Maker has a strategy such that after each full Maker–Breaker round Q is at least its value before Maker's move. Consequently, if the game is continued until no live positive-weight edge remains, the final scored weight is at least

$$\frac{1}{8} \sum_{e \in \mathcal{E}} w(e).$$

The same conclusion holds for the rank-2 graph variant in which Breaker may delete an unscored live edge instead of scoring it, and Maker-scored edges are called claimed edges.

Proof. We prove the rank-3 scored statement; the graph version is the same argument with edge deletion replacing Breaker's scored-edge move. The graph edge deletion loses exactly the current live potential of that edge, while a scored-edge move changes Q by $8w(e) - \Phi(e) \geq 0$.

For a live Maker move f, let

$$\Delta(f) := Q(\text{after Maker scores } f) - Q(\text{before Maker scores } f).$$

Maker chooses a live edge maximizing $\Delta(f)$. Such a choice exists because the game is finite.

We record the two local domination facts used in the round estimate. First, if g is live and $|g \cap C| \le 2$, then scoring g itself increases Q by at least its current live potential $\Phi(g)$. Indeed, the contribution of g changes from $2^r w(g)$ to $8w(g)$ with $r = |g \cap C| \le 2$, and $8w(g) - 2^r w(g) \ge 2^r w(g)$. All other live-edge contributions can only increase when Maker captures new vertices.

Second, let v be a vertex that Breaker could delete after Maker's move, and let Λ_v be the live potential incident with v at that moment. If $\Lambda_v > 0$, choose a live edge g containing v after Maker's move. The edge g was also live before Maker's move. Compare Maker's actual max-gain edge with the alternative move g. When Maker scores g, every live edge through v at the beginning of the round gains its whole current potential from the capture of v, and the selected edge g supplies enough additional gain to cover the possible coefficient increases caused by the actual move before v is deleted. Edge by edge, this gives

$$\Delta(g) \ge \Lambda_v.$$

Since Maker chose an edge of maximum Δ, the actual Maker gain is at least Λ_v.

Now consider Breaker's reply. If Breaker scores a live edge, Q does not decrease, as observed above. If Breaker deletes an uncaptured vertex v, the only lost live terms are the live edges incident with v, whose total potential is Λ_v after Maker's move. The previous paragraph shows that Maker's already-secured gain is at least Λ_v, so the full Maker–Breaker round does not decrease Q.

Initially $C = \varnothing$, $S = 0$, and every edge is live, so

$$Q_0 = \sum_{e \in \mathcal{E}} w(e).$$

At terminal time there is no positive-weight live edge, hence $Q_{\text{fin}} = 8S_{\text{fin}}$. Monotonicity gives $8S_{\text{fin}} \ge Q_0$, which is the desired one-eighth bound. □

Proposition A.2 (Initial target mass)**.** *For every fixed* $0 < \delta < 1/4$*,*

$$W_0 := \sum_{\substack{a<c \\ a,c \in \mathcal{P}_Y}} |B_{a,c}| \gg_\delta \frac{n(\log\log n)^2}{\log n}.$$

Proof. Since $ac \le Y^2 = n^{2\delta}$, the interval endpoint $X_{a,c} := n/(ac)$ satisfies $X_{a,c} \ge n^{1-2\delta} \to \infty$. By the prime number theorem, uniformly for all such pairs and all sufficiently large n,

$$\pi(X_{a,c}) - \pi(X_{a,c}/2) \ge \kappa_\delta \frac{X_{a,c}}{\log n} = \kappa_\delta \frac{n}{ac \log n}$$

for some positive constant κ_δ. Therefore

$$W_0 \ge \kappa_\delta \frac{n}{\log n} \sum_{\substack{a<c \\ a,c \in \mathcal{P}_Y}} \frac{1}{ac}.$$

Mertens' theorem for primes gives

$$\sum_{p \in \mathcal{P}_Y} \frac{1}{p} = \log\log Y + O(1) = \log\log n + O_\delta(1),$$

and $\sum_p p^{-2} < \infty$. Hence

$$\sum_{a<c \in \mathcal{P}_Y} \frac{1}{ac} = \frac{1}{2}\left[\left(\sum_{p \in \mathcal{P}_Y} \frac{1}{p}\right)^2 - \sum_{p \in \mathcal{P}_Y} \frac{1}{p^2}\right] = \left(\frac{1}{2} + o(1)\right)(\log\log n)^2.$$

Combining the last two displays proves the claim. □

Proposition A.3 (Residual divisibility embedding)**.** *Let $\mathcal{T}_*$ be any subfamily of targets $t = acb \in \mathcal{T}$ such that, for every $t = acb \in \mathcal{T}_*$, the numbers a, c, and ac are already unavailable as legal moves. Attach to t the three slot labels*

$$b, \qquad ab, \qquad cb$$

and let $e_t := \{b, ab, cb\}$. Then the residual divisibility game on $\mathcal{T}_*$ is at least as favorable to Prolonger as the scored rank-3 hypergraph game on the hyperedges $\{e_t : t \in \mathcal{T}_*\}$:

1. *the only future moves that can make $t = acb$ illegal are b, ab, cb, and t itself;*
2. *playing b, ab, or cb deletes exactly the residual targets whose hyperedges contain that slot;*
3. *playing the exact target t kills no other target in $\mathcal{T}_*$, and is therefore a scored-edge move;*
4. *every live hyperedge corresponds to a legal actual move t.*

Proof. Fix $t = acb \in \mathcal{T}_*$. Because $t > n/2$, no proper multiple of t lies in $\{2, \dots, n\}$. Thus a future move can make t illegal only by being t itself or a proper divisor of t. The positive divisors are

$$1,\ a,\ c,\ ac,\ b,\ ab,\ cb,\ acb.$$

The divisor 1 is not a move, while a, c, and ac are unavailable by hypothesis. This proves the first assertion.

For slot incidence, suppose another target is $t' = a'c'b'$. Since $b, b' > Y \geq a, a', c, c'$, the divisibility $b \mid t'$ holds iff $b = b'$. Thus a move b kills exactly the targets in the b-fiber. Similarly, if $ab \mid a'c'b'$, the large prime factor forces $b = b'$, and after cancellation a must be a' or c'. This is exactly the condition that the slot ab belongs to $e_{t'}$. The proof for cb is identical.

If two targets t, u in $\mathcal{T}_*$ are comparable, say $t \mid u$, then u is a multiple of t lying in $(n/2, n]$. The next positive multiple of t after t is $2t > n$, so $u = t$. Hence distinct targets are pairwise incomparable, and playing an exact target scores only that target.

Finally, if e_t is live in the slot hypergraph, then none of b, ab, cb, or t has been played. The only other proper divisors of t are $1, a, c, ac$, and these are either not moves or already unavailable. There is no proper multiple of t on the board. Therefore no played number is comparable with t, so the actual move t is legal. This proves the comparison. □

Proposition A.4 (Activation-stage potential)**.** *During activation, let each unclaimed pair edge $e = (a, c)$ carry the current token set $B_e(t)$ of primes $b \in B_{a,c}$ for which the target acb has not yet been killed. Write $w_t(e) := |B_e(t)|$.*

Prolonger plays as follows. In each activation round, among pair edges with positive current token weight, he chooses a max-gain edge in the graph version of Proposition A.1; if $e = (a, c)$ is chosen, he plays one live target acb with $b \in B_e(t)$. This scores one actual move, claims the pair edge e, captures the vertices a and c, and leaves the remaining live tokens on that secured pair for the residual phase.

Let S_t be the number of activation targets already played by Prolonger, and define

$$Q_t := S_t + \sum_e \phi_t(e),$$

where

$$\phi_t(e) = \begin{cases} \frac{1}{8} w_t(e), & e \text{ unclaimed and neither endpoint captured}, \\ \frac{1}{4} w_t(e), & e \text{ unclaimed and one endpoint captured}, \\ \frac{1}{2} w_t(e), & e \text{ unclaimed and two endpoints captured}, \\ w_t(e), & e \text{ claimed}, \\ 0, & e \text{ deleted}. \end{cases}$$

Let E be the total number of target tokens deleted during activation by Shortener moves not modeled as graph vertex deletions a or graph edge deletions ac. At the end of activation,

$$Q_{\text{end}} \geq \frac{W_0}{8} - E.$$

Equivalently,

$$S_{\text{end}} + \sum_{e \text{ secured}} w_{\text{end}}(e) \geq \frac{W_0}{8} - E.$$

Proof. First ignore off-model token deletions. The activation process is exactly the rank-2 graph game of Proposition A.1, with one harmless bookkeeping change: when Prolonger claims an edge e, he consumes one token from e and adds it immediately to the score S_t. If the coefficient of e before the move is $\gamma \in \{1/8, 1/4, 1/2\}$, the contribution of e changes from $\gamma w_t(e)$ to

$$1 + (w_t(e) - 1) = w_t(e),$$

exactly as if the edge had become claimed with full weight. All other incident edges change as in the graph game. Thus the scaled potential argument gives $Q_{\text{end}} \geq W_0/8$ when every harmful Shortener move is a graph deletion.

Now restore all actual Shortener moves. A token acb is deleted precisely when Shortener plays one of its still-available harmful divisors or the target itself. Deleting one live token can reduce Q_t by at most 1, since every coefficient appearing in the definition of $\phi_t(e)$ is at most 1. Therefore all off-model deletions together reduce the lower bound by at most E, giving

$$Q_{\text{end}} \geq W_0/8 - E.$$

At the end of activation, a positive-weight edge is either secured or deleted, so Q_{end} is exactly the activation score plus the residual live target weight on secured pairs. □

Proposition A.5 (Activation deletion budget)**.** *For fixed $0 < \delta < 1/4$, the off-model deletion count from Proposition A.4 satisfies*

$$E \ll \frac{Y^4}{\log^4 Y} = o\left(\frac{n(\log\log n)^2}{\log n}\right).$$

Proof. There is at most one activation round per small-prime pair, so the number of rounds is

$$R \leq \binom{|\mathcal{P}_Y|}{2} \ll \frac{Y^2}{\log^2 Y}.$$

We bound the number of live tokens an off-model Shortener move can delete.

If Shortener plays a large prime b, then for each pair (a, c) there is at most one token acb using that b. Hence the move deletes at most $O(Y^2/\log^2 Y)$ tokens. If Shortener plays a lateral semiprime pb with $p \leq Y$ and $b > Y$ prime, the killed targets have the form pcb with the third small prime varying, so there are at most $O(Y/\log Y)$ of them. If Shortener plays an exact target acb, exactly one token is deleted.

The worst of these bounds is the large-prime bound. Across R activation rounds,

$$E \ll R\frac{Y^2}{\log^2 Y} \ll \frac{Y^4}{\log^4 Y}.$$

Since $Y = n^\delta$,

$$\frac{Y^4/\log^4 Y}{n(\log\log n)^2/\log n} \ll_\delta \frac{n^{4\delta-1}}{(\log n)^3(\log\log n)^2} \to 0$$

because $4\delta - 1 < 0$. □

Proposition A.6 (Residual mass on secured pairs)**.** *At the end of activation, the secured pairs carry residual live target weight*

$$M := \sum_{e \text{ secured}} w_{\text{end}}(e) \gg_\delta \frac{n(\log\log n)^2}{\log n}.$$

Proof. By Proposition A.4,

$$S_{\text{end}} + M \geq \frac{W_0}{8} - E.$$

The activation score is at most the number of small-prime pairs:

$$S_{\text{end}} \leq R \ll \frac{Y^2}{\log^2 Y} = o\left(\frac{n(\log\log n)^2}{\log n}\right),$$

because $2\delta < 1$. The deletion budget E is negligible by Proposition A.5, while W_0 has the required lower bound by Proposition A.2. Subtracting the two negligible terms gives the claim. □

Proposition A.7 (T2 lower bound)**.** *For every fixed $0 < \delta < 1/4$ there is $c_\delta > 0$ such that*

$$L(n) \geq c_\delta \frac{n(\log\log n)^2}{\log n}$$

for all sufficiently large n.

Proof. After activation, restrict to the residual live targets lying over secured pairs. Their total weight is M, as in Proposition A.6. For each secured pair (a, c), the activation move made a, c, and ac unavailable. Therefore Proposition A.3 compares the remaining divisibility game to a scored rank-3 slot hypergraph whose total edge weight is M.

By Proposition A.1, Prolonger can force scored residual target weight at least $M/8$ in this hypergraph model. The comparison proposition says that every such scored hypergraph move is a genuine actual game move, and that the actual game gives Prolonger no fewer legal options than the model. Hence the number of residual moves after activation is at least $M/8$. Since $M \gg_\delta n(\log\log n)^2/\log n$, the same lower bound holds for the total game length.

Choosing any fixed $\delta < 1/4$, for instance $\delta = 1/8$, gives an absolute constant $c > 0$ and proves Theorem 3.4. □

Appendix B. Numerical Computations

This appendix records the reproducible numerical checks that accompany the proofs in Sections 6 and 8. They are not used as substitutes for proof, but they do certify the quoted constants and stress-test the finite combinatorial steps.

B.1. **Bonferroni–4 Constant.**

Theorem 6.5 uses the finite Bonferroni–4 constant

$$\frac{\mathcal{W}_4}{2} = 0.1897112 < 0.19.$$

Two numerical artifacts accompany this value.

(1) The direct grid computation in `phase3/shortener_piecewise_bonferroni4_summary.md`, whose finest-grid output is

$$\frac{\mathcal{W}_4}{2} = 0.189710592;$$

(2) the Round 57/60 reconciliation in `r57_bonferroni4_audit_and_repair.md` and `researcher-60-pro-R57 md`, which gives the paper-facing theorem constant

$$\frac{\mathcal{W}_4}{2} = 0.1897112.$$

The seventh-decimal discrepancy is benign: the theorem only needs a strict margin below 0.19, and both computations provide that margin comfortably.

Grid	J_1	J_2	J_3	J_4	$\mathcal{W}_4$	$\mathcal{W}_4/2$
2^{13}	0.78844185	0.18676003	0.02008041	0.00122154	0.37945930	0.18972965
2^{14}	0.78849100	0.18679383	0.02008737	0.00122220	0.37943767	0.18971883
2^{15}	0.78851725	0.18681153	0.02009097	0.00122255	0.37942586	0.18971293
2^{16}	0.78852127	0.18681405	0.02009149	0.00122260	0.37942388	0.18971194
2^{17}	0.78852704	0.18681777	0.02009220	0.00122266	0.37942118	0.18971059

TABLE 2. Grid convergence for the Bonferroni–4 truncation.

B.2. **Round 60 Sandbox Checks.** The R60 repair included four targeted finite checks:

(1) *Odd-part injection.* Exhaustive verification at $n = 14$ over all 732 antichains; no collision of odd parts occurs.
(2) *Monotone replacement.* Exact inclusion–exclusion counts at $n = 10^4, 10^5, 10^6$ for several test pairs (q_j) and (p_j) with $q_j \le p_j$, always confirming $N(q_1, \dots, q_K) \le N(p_1, \dots, p_K)$.
(3) *Squarefree divisor counts.* For $r \le 4$, the counts $M_r(n)$ of relevant squarefree r-fold divisors match the scale $O_r(n(\log\log n)^{r-1}/\log n)$ used in the Bonferroni floor-error bound.
(4) *Controlled prime rounding.* Synthetic monotone b-prefixes were rounded upward to increasing odd-prime prefixes at $n = 10^4, 10^5, 10^6, 10^8$; the resulting changes in the moment terms were small and always moved in the monotone direction predicted by the proof.

Check	Recorded outcome
Odd-part injection	`(True, None, None, 732)` at $n = 14$
Monotone replacement	Verified at $n = 10^4, 10^5, 10^6$ for all listed test prefixes
M_r scaling	Ratified for $r = 1, 2, 3, 4$ up to $n = 10^6$
Prime-rounding bridge	Tested on synthetic prefixes up to $n = 10^8$

TABLE 3. Finite checks recorded in the R60 audit-repair note.

B.3. **What the Numerics Do and Do Not Say.** The numerical artifacts support three distinct claims:

(1) the quoted theorem constant $\mathcal{W}_4/2 < 0.19$ is reproducible;
(2) the finite steps used in the proof of Theorems 6.3 and 6.4 behave as expected in representative regimes;
(3) the sharper full-series value near 0.18969 is numerically stable.

They do *not* replace the analytic arguments of Section 6. In particular, the theorem statement in the paper is the finite Bonferroni–4 bound below 0.19, not the sharper extrapolated constant from the full alternating series.

APPENDIX C. FORMALIZATION DETAILS

This appendix records the current explicit Lean gaps in the checked-in files. The point is not to reproduce the proofs from Section 10, but to make the remaining obligations auditable. All paths below are relative to the public artifact repository [Bud26], with the Lean subtree rooted at `erdos-872/lean/`.

C.1. **Closed Projects.** The following projects compile with no remaining `sorry` terms in their main mathematical files:

- `shield_reduction/shield_reduction_aristotle/RequestProject/ShieldReduction.lean`;
- `tau_5_24/tau_5_24_aristotle/RequestProject/\{Cover,Packing,Tau\}.lean`;
- `shortener_13_36/shortener_13_36_v2_aristotle/RequestProject/Shortener/MainTheorem.lean`;
- `erdos_872_core/RequestProject/T2Finite/*.lean`;

- `erdos_872_core/RequestProject/Round15Bonferroni4/\{Envelope,FlatMass,Inversion,PrimeBounds,Target\}.lean`.

For the last two bullets, "zero-sorry" means that the currently formalized reductions are closed; it does not mean that the full prose theorems of T2 or the full analytic Round 15 proof have been formalized end-to-end.

C.2. **Explicit `sorry` Inventory.**
Theorem A / T1. The file `theorem_A/theorem_A_shield_lower_bound_aristotle/RequestProject/ShieldMainTheorem.lean` contains the following five explicit `sorry` sites.

(1) Line 50: `mertens_second_theorem`. Category: analytic number theory / Mertens' theorem.
(2) Line 55: `chebyshev_upper_bound`. Category: prime-counting upper bound.
(3) Line 63: `chebyshev_theta_asymptotic`. Category: Chebyshev-ϑ / PNT-adjacent asymptotics.
(4) Line 217: `shield_lower_bound`. Category: final theorem statement depending on the three analytic inputs above.
(5) Line 224: `barrier_exponent`. Category: corollary depending on the main theorem.

In other words, the combinatorial shielding core is formalized; the unresolved part is the analytic prime-sum infrastructure.
5/16 upper bound. The file `shortener_5_16/shortener_5_16_aristotle/RequestProject/Shortener516/Theorems.lean` contains one remaining explicit `sorry`:

(1) Line 499: `game_value_per_parameter`. Category: bundled analytic / game-tree gap. As documented in the file, this single placeholder packages two missing ingredients:
  - a Chebyshev-type theorem supplying the needed prime prefix estimate;
  - the game-tree wrapper showing that Shortener can realize the prescribed legal odd-prime prefix for the requisite number of turns.

All preceding combinatorial compression statements in that file are zero-sorry.

C.3. **Interpretation.** The formalization status is strongest for the static reductions and the intermediate 13/36 strategy theorem. The remaining formal gaps are of two types only:

(1) classical analytic inputs not yet exposed conveniently in the pinned Mathlib snapshot;
(2) wrappers translating a formally verified finite core into the exact asymptotic theorem stated in the paper.

This is why the paper treats Theorems 2.2 and 5.1 and the Round 15 endgame reduction as fully formalized, treats the structural core of Theorem 4.2 as fully formalized with an elementary cardinality post-processing step, and presents Theorems 3.1, 3.4, 5.3 and 6.5 with more precise caveats.

Appendix D. Glossary and Notation

D.1. **Game Notation.**

$L(n)$**:** Length of the divisibility antichain game on $\{2, \dots, n\}$ when both players play optimally and Prolonger moves first.
U**:** The upper half $(n/2, n] \cap \mathbb{Z}$ of the board. Any subset of U is an antichain.
A**:** A terminal antichain, or more generally the current played set.
P**:** A set of upper-half Prolonger moves, usually a shield set in the Shield Reduction framework.
$M_n(x)$**:** The shadow of x in the upper half: $M_n(x) = \{u \in U : x \mid u\}$.
$w_n(x)$**:** The shadow weight $|M_n(x)|$.
$\beta(P)$**:** The maximum total weight $\sum_{x \in B} w_n(x)$ over antichains $B \subseteq L(P)$ in the lower half; the largest weighted lower-half antichain surviving after Prolonger plays P.

D.2. Lower-Bound Notation.

Theorem A / T1: The polynomial shield lower bound and its corollary on the $n^{1/e}$ barrier for short shield prefixes.

T2: The Maker-first hypergraph-capture lower bound at scale $n(\log\log n)^2/\log n$.

$r_1(n)$**:** The scale $n(\log\log n)^2/\log n$, used repeatedly in the later upper-bound and barrier discussions.

D.3. Upper-Bound Notation.

σ_{15}**:** The Round 15 Shortener strategy: play the smallest legal odd prime during the first $K = \lfloor (1-\varepsilon)n/(2\log n)\rfloor$ turns.

$q_1 < \cdots < q_K$**:** The odd primes actually claimed by Shortener during the σ_{15} prefix.

$\rho(u)$**:** The piecewise density of the Round 15 prefix, namely $\rho(u) = 1/((\lfloor 1/u\rfloor + 1)u)$ away from the breakpoints $u = 1/h$.

J_r**:** The rth alternating-convolution moment attached to ρ.

$\mathcal{W}_4$**:** The finite Bonferroni–4 constant $1 - J_1 + J_2 - J_3 + J_4$.

$\mathcal{W}$**:** The full alternating series $\sum_{r\geq 0}(-1)^r J_r$, used only numerically in this paper.

b_j**:** The monotone comparison scale obtained from the Round 57 envelope inversion.

p_j**:** The increasing odd-prime sequence used in the prime-rounding bridge from b_j to the Bonferroni comparison theorem.

D.4. Barrier Notation.

σ^***:** The max-unresolved-harmonic-degree Shortener strategy studied in the packet-era notes. It is distinct from σ_{15}.

τ_{SF}**:** The separator-first / prime-fallback strategy appearing in the late separator program.

$\tau_{\mathbb{Z}}(\mathcal{C})$**,** $\tau_f(\mathcal{C})$**:** Integral and fractional transversal numbers of a uniform family $\mathcal{C}$.

$\mathrm{SA}_r(\mathcal{C})$**:** Level-$r$ Sherali–Adams relaxation value for the same covering instance.

δ_q**,** λ_q^2**:** The capture density and Johnson-scheme spectral parameter in the fixed-rank q-shadow dichotomy.

$\sigma_q(\mathcal{D})$**:** The fraction of the q-layer already occupied by the blocker shadow of $\mathcal{D}$.

Artifact Availability

The public artifact repository for this paper is [Bud26]. It contains the Lean formalizations, Aristotle submission files and outputs, numerical scripts, and supporting research notes used in the preparation of the present manuscript. Repository-relative paths quoted in Section 10 and appendices B and C are relative to that artifact root.

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Declaration of Generative AI and AI-assisted Technologies in the Research and Writing Process

During the preparation of this work, the author used generative AI systems including OpenAI GPT-5.4 Pro, Anthropic Claude Opus 4.7, and Google Gemini 3.1 DeepThink for exploratory discussion, adversarial proof auditing, code and Lean scaffolding, numerical cross-checking, artifact organization, and drafting assistance. After using these tools, the author reviewed, corrected, and rewrote the material as needed, verified references and formal artifacts against the repository, and takes full responsibility for the claims, proofs, citations, and presentation in the final manuscript.

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